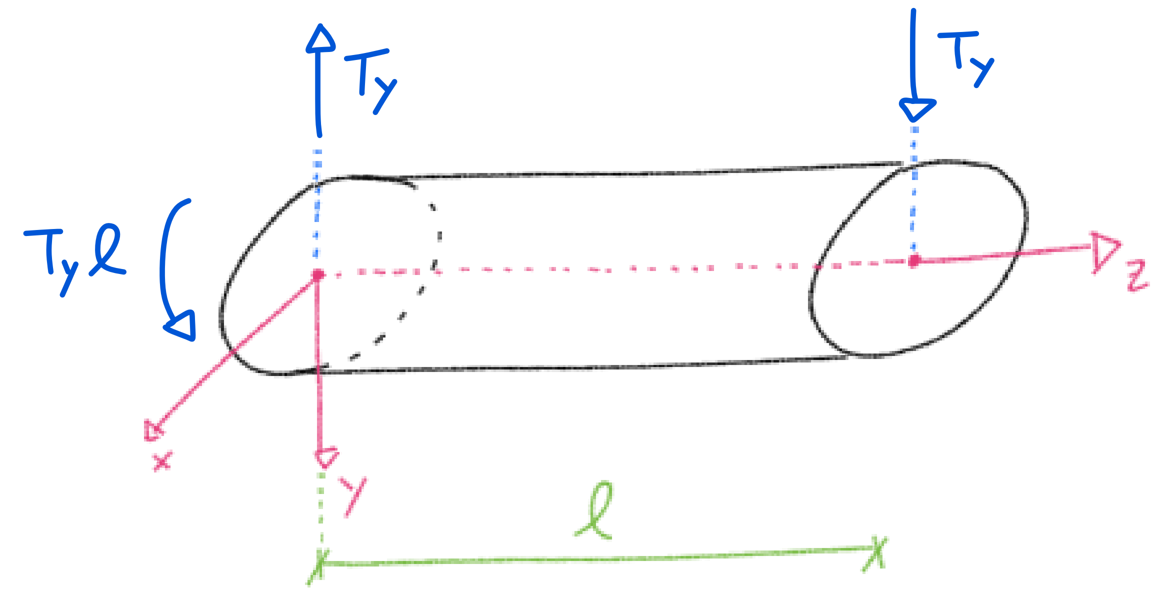
Corso di
Scienza delle Costruzioni

IL PROBLEMA DI SAINT VENANT
FLESSIONE E TAGLIO

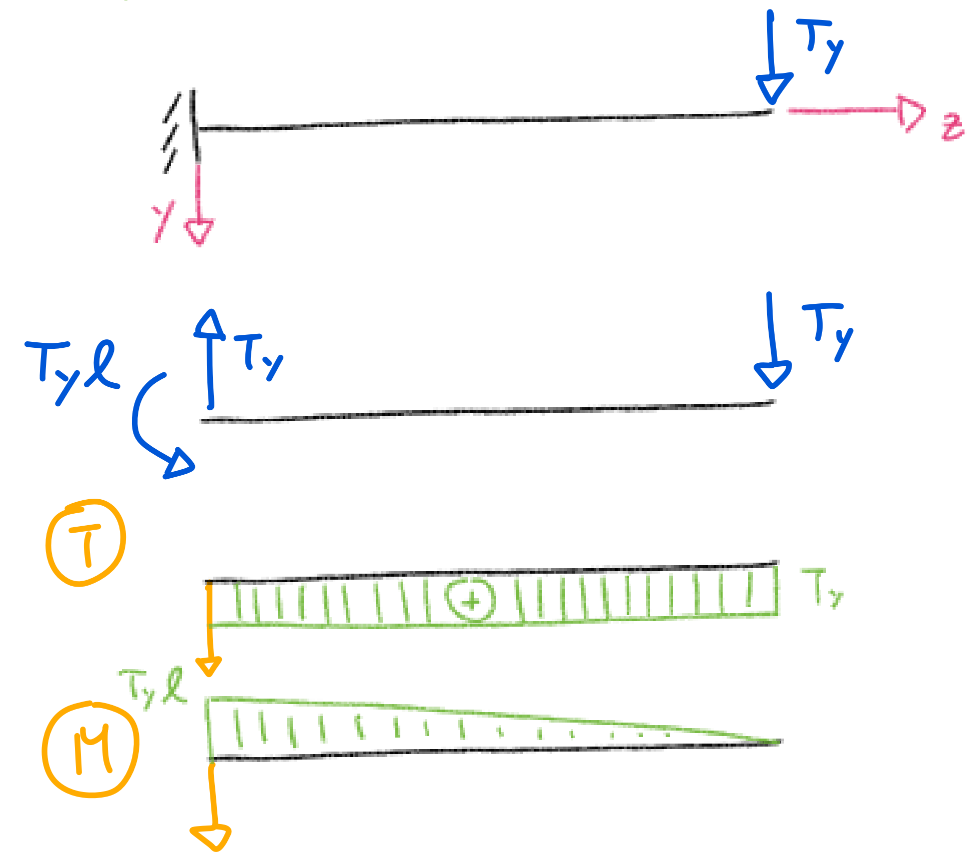
TAGLIO RETTO

T_y LUNGO ASSE PRINC. INERZIA

$$\underline{\underline{T}} : \begin{bmatrix} 0 & 0 & \tau_{zx} \\ 0 & 0 & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix}$$



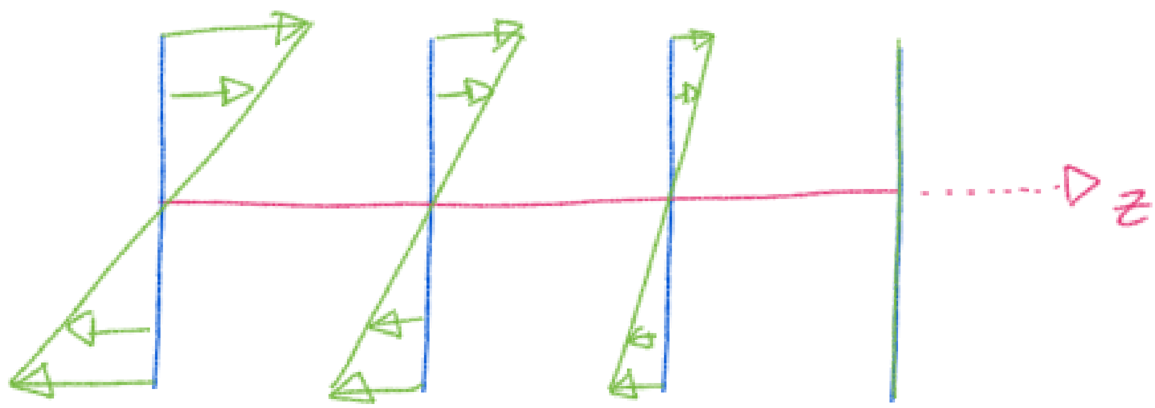
CdS: $M_x = -T_y(\ell - z)$ $\left\{ \begin{array}{l} \text{LINEARE} \\ \text{IN } z \end{array} \right.$



1) SFORZI NORMALI (Navier)

$$\sigma_z = \frac{M_x}{I_x} y = -\frac{T_y(\ell - z)}{I_x} y$$

OGNI SEZIONE
CON UNA
DISTRIBUZIONE
DIVERSA



SFORZI TANG.

TAGLIO RETTO: TENSIONI TANGENZIALI

EQ. IND. DI EQUIL.: $\nabla \cdot \underline{\underline{T}} + \underline{b} = \underline{0}$

$$\underline{\underline{T}} = \begin{bmatrix} 0 & 0 & \tau_{zx} \\ 0 & 0 & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \Rightarrow \operatorname{div}(\underline{\underline{T}}) = \underline{0}\,?$$

$\underline{\tau} = [\tau_{zx}, \tau_{zy}]^T$

$$\operatorname{div}(\underline{\underline{T}}) = \frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{zy}}{\partial y} + \frac{\partial \sigma_z}{\partial z} \equiv \operatorname{div}(\underline{\tau}) + \frac{\partial \sigma_z}{\partial z}$$

$$\sigma_z = -\frac{T_y(\ell - z)}{I_x}\, y \Rightarrow \frac{\partial \sigma_z}{\partial z} = \frac{T_y}{I_x}\, y$$

$$\Rightarrow \operatorname{div}(\underline{\tau}) = -\frac{T_y}{I_x}\, y \neq 0$$

CAMPO NON SOLENOIDALE

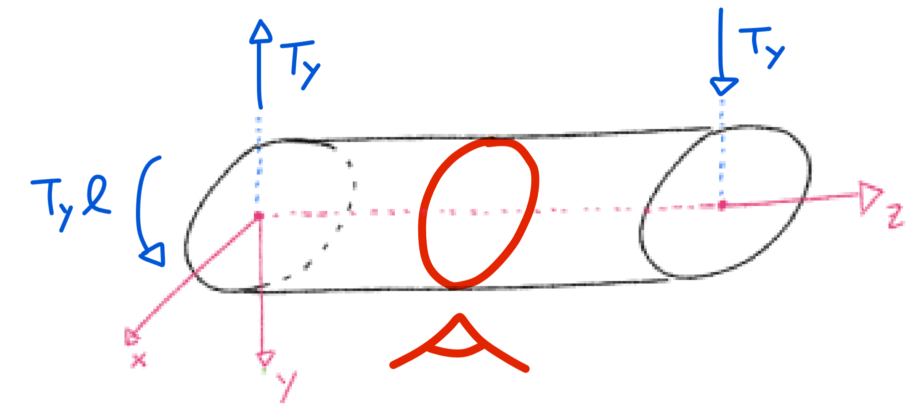
(LINEE APERTE)

TAGLIO RETTO: TENSIONI TANGENZIALI ALLA JOURAWSKY

Hp: · SEZIONI DI PICCOLO SPESSORE

· PRINCIPIO DI REC. T.T. :

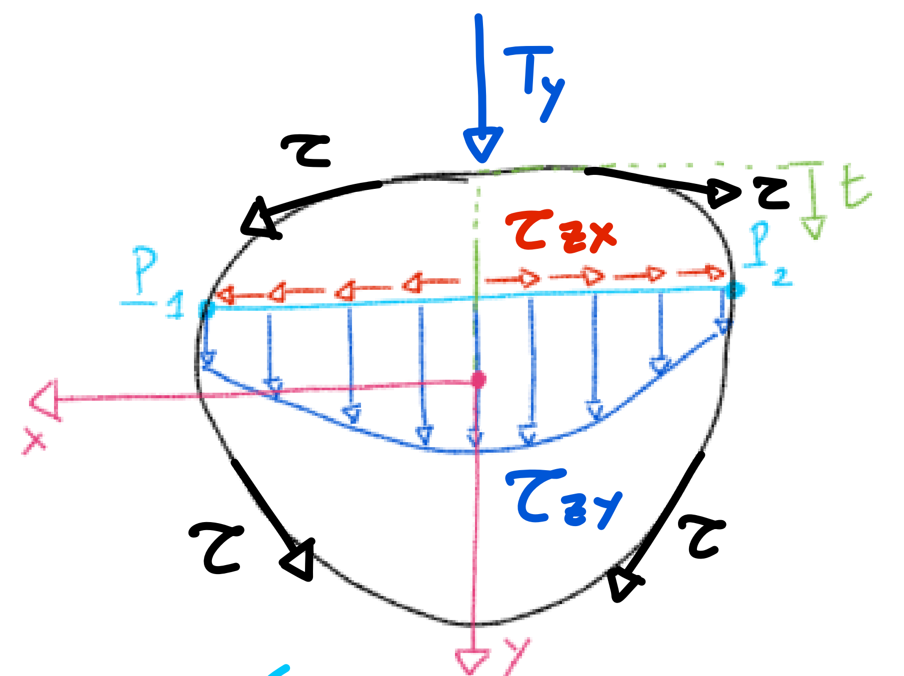
$\hookrightarrow \quad \underline{\tau} = \tau_{zx}\,\hat{i} + \tau_{zy}\,\hat{j}$



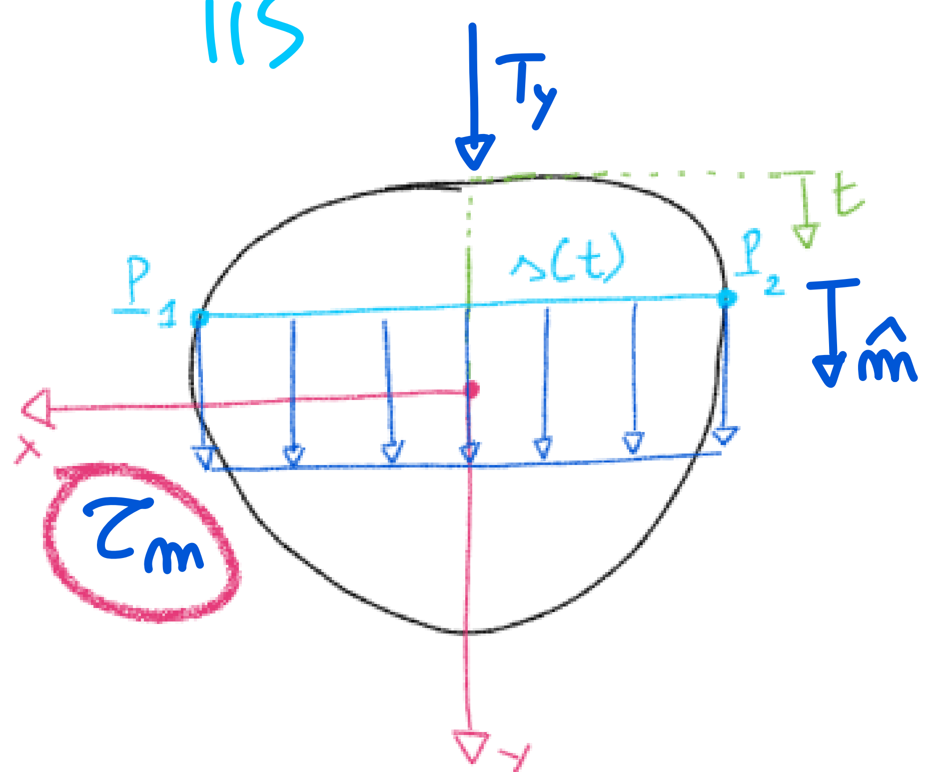
JOURAWSKY : $\overline{P_1P_2} = s(t)$, $\hat{m}$ USCENTE

$$\left.\begin{aligned} \tau_{zx} &= f_1(x,y) \\ \tau_{zy} &= f_2(x,y) \end{aligned}\right\}$$

LE τ SONO INDIPENDENTI DA z QUINDI SONO IDENTICHE IN OGNI SEZIONE ✱ ✱



||S



Hp. : APPROX. CON VALORE MEDIO RISPETTO ASSE DI SIMMETRIA (Y)

✱ UGUAGLIANZA DELLE AREE

$\hookrightarrow \quad \tau_{zx} \Rightarrow 0 \qquad (\text{SIMM. } y \Rightarrow \Sigma = 0)$

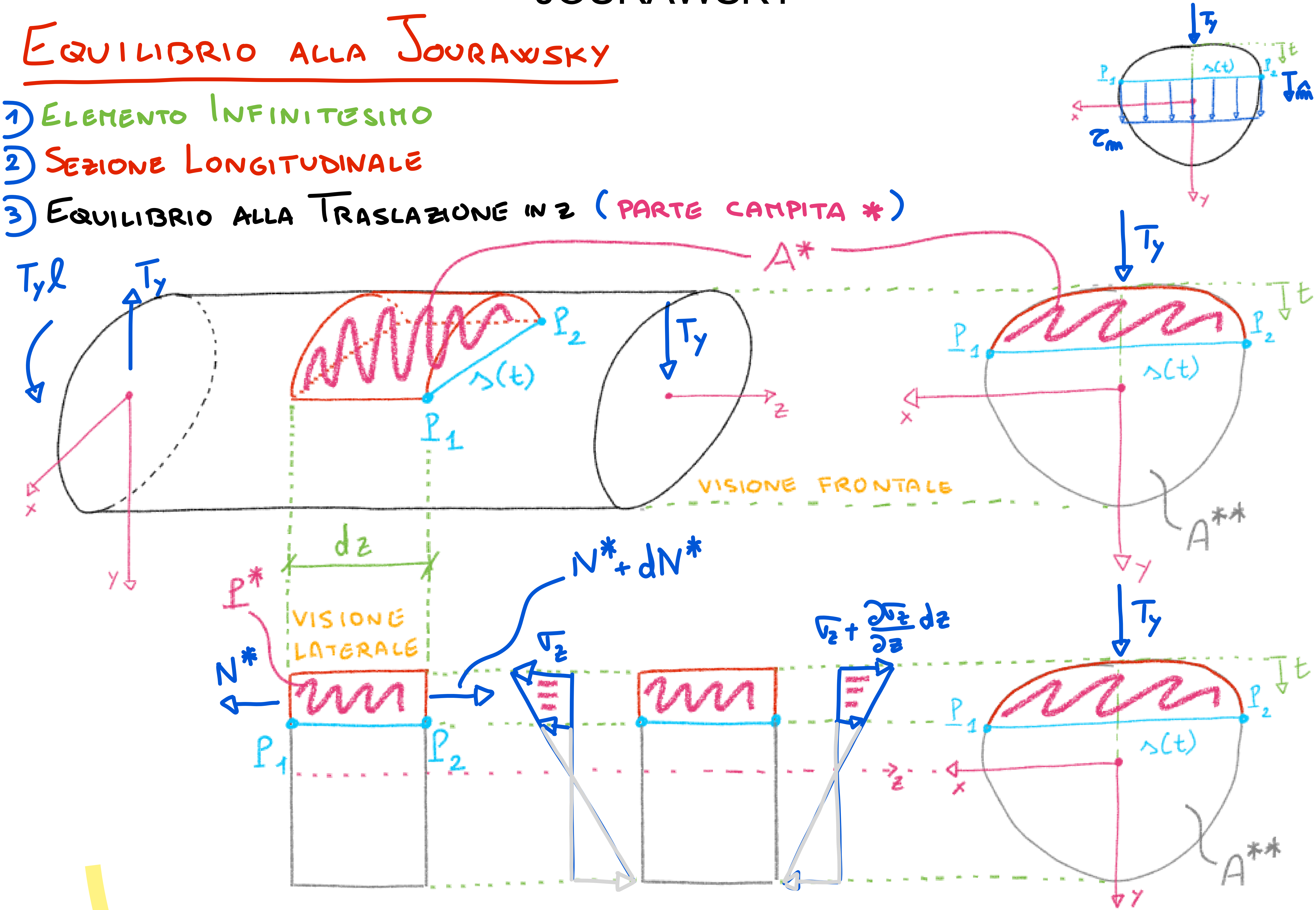
$\hookrightarrow \quad \tau_{zy} \Rightarrow \tau_m = \dfrac{1}{s(t)} \displaystyle\int_{-s(t)/2}^{s(t)/2} \tau_{zy}\, dx$

?

TAGLIO RETTO: TENSIONI TANGENZIALI ALLA JOURAWSKY

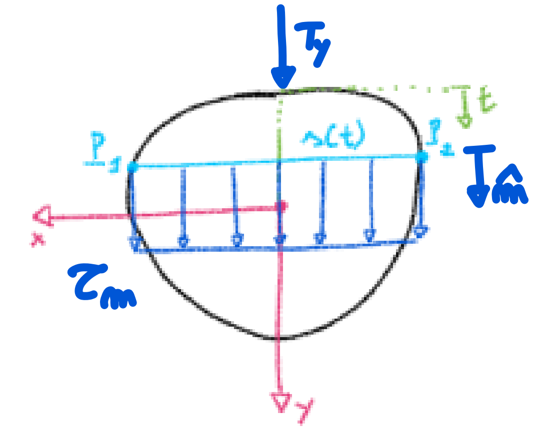
EQUILIBRIO ALLA JOURAWSKY

1) ELEMENTO INFINITESIMO
2) SEZIONE LONGITUDINALE
3) EQUILIBRIO ALLA TRASLAZIONE IN z (PARTE CAMPITA *)



TAGLIO RETTO: TENSIONI TANGENZIALI ALLA JOURAWSKY

EQUILIBRIO ALLA JOURAWSKY



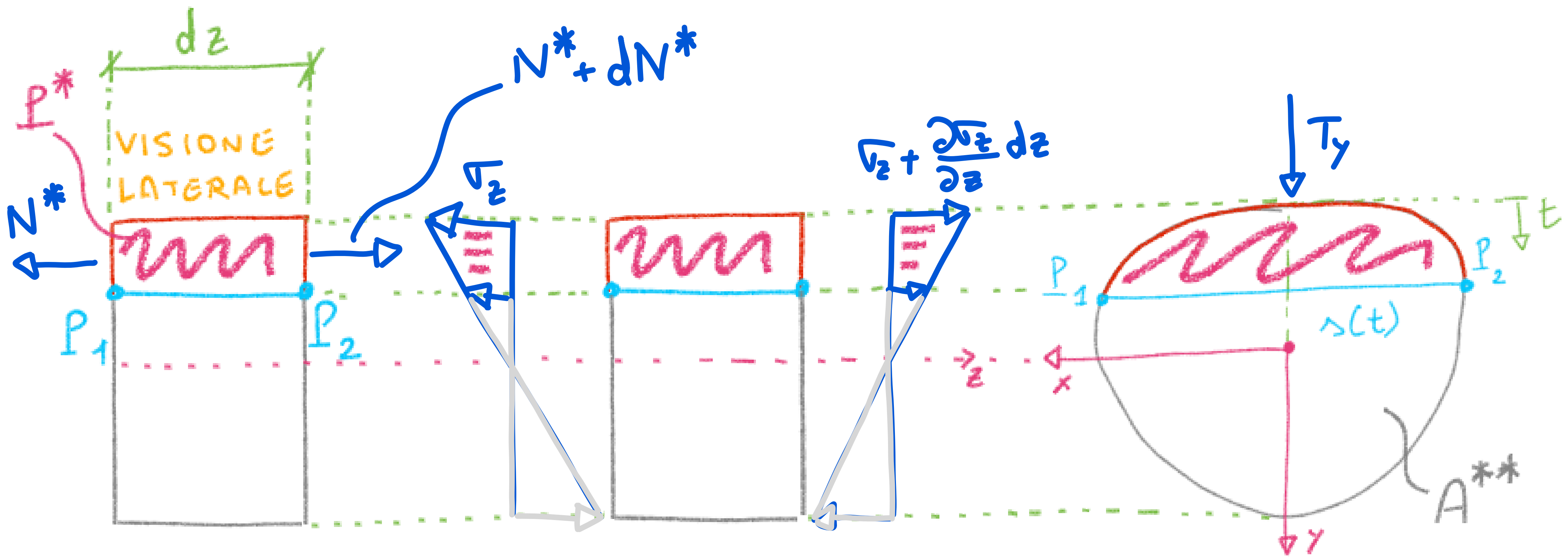
1) ELEMENTO INFINITESIMO
2) SEZIONE LONGITUDINALE
3) EQUILIBRIO ALLA TRASLAZIONE IN z (PARTE CAMPITA *)

RISULTANTI SFORZI NORMALI:

$$N^* = \int_{A^*} \sigma_z \, dA^* \; ; \qquad N^* + dN^* = \int_{A^*} \left(\sigma_z + \frac{\partial \sigma_z}{\partial z} dz \right) dA^*$$

$$\sigma_z = - \frac{T_y (\ell - z)}{I_x} y$$

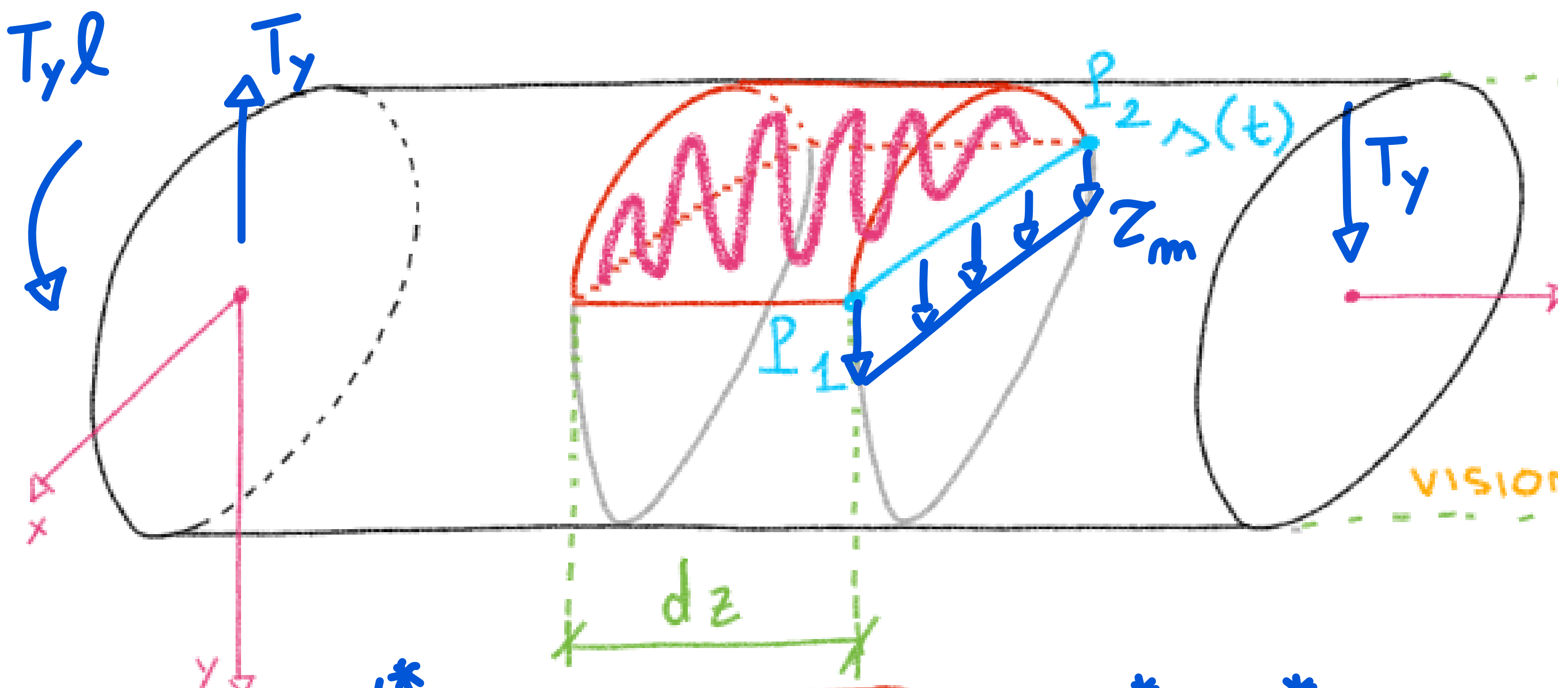
$$dN^* = \int_{A^*} \frac{\partial \sigma_z}{\partial z} dz \, dA^* = \int_{A^*} \frac{T_y}{I_x} y \, dz \, dA^* = \frac{T_y}{I_x} dz \int_{A^*} y \, dA^* = \frac{T_y}{I_x} S_x^* \, dz$$

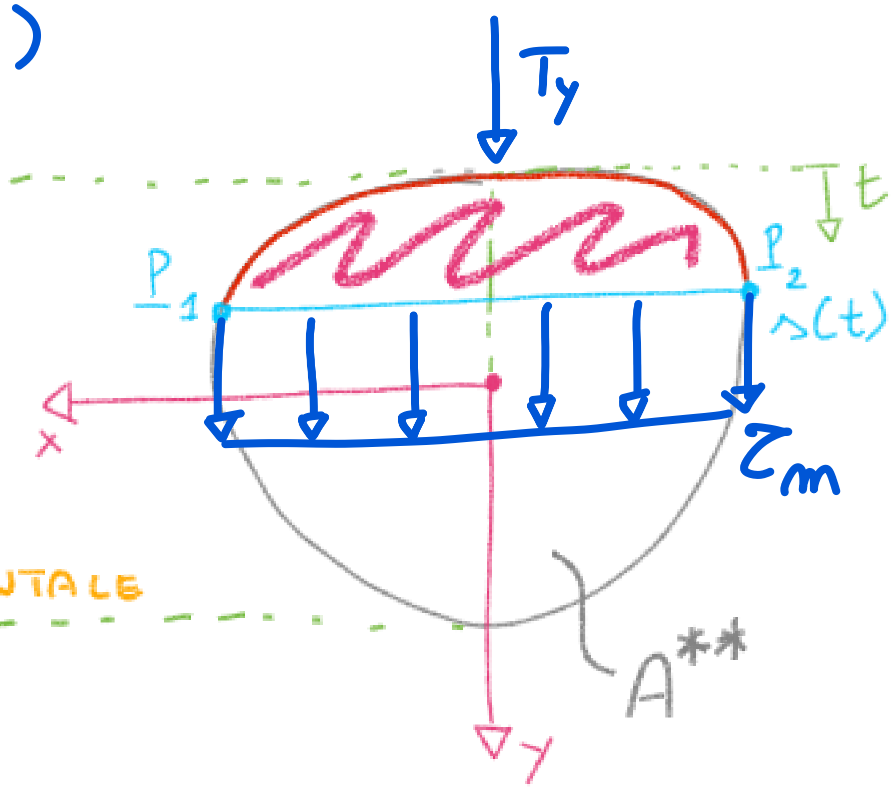


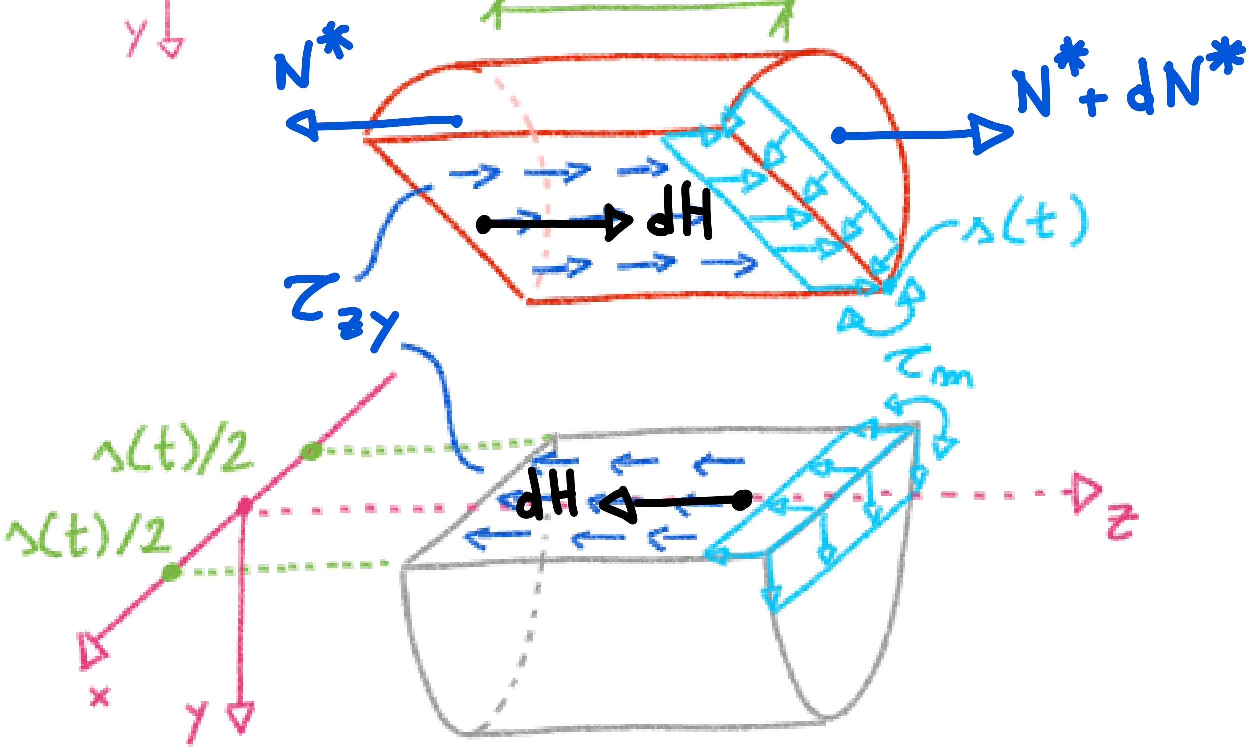
TAGLIO RETTO: TENSIONI TANGENZIALI ALLA JOURAWSKY

EQUILIBRIO ALLA JOURAWSKY $\Longrightarrow$ CIOCCOLATINO PERUGINA

1) ELEMENTO INFINITESIMO
2) SEZIONE LONGITUDINALE
3) EQUILIBRIO ALLA TRASLAZIONE IN z (PARTE CAMPITA *)







RISULTANTE SFORZI TANGENZIALI

$$dH = \left(\int_{-s(t)/2}^{s(t)/2} \tau_{zy} \, dx \right) dz \doteq s(t) \, \tau_m \, dz$$

$$\tau_m = \frac{1}{s(t)} \int_{-s(t)/2}^{s(t)/2} \tau_{zy} \, dx$$

TAGLIO RETTO: TENSIONI TANGENZIALI ALLA JOURAWSKY

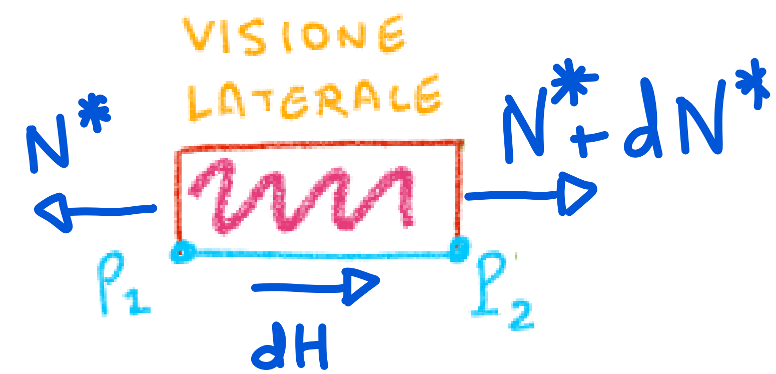
EQUILIBRIO ALLA JOURAWSKY $\Longrightarrow$ CIOCCOLATINO PERUGINA

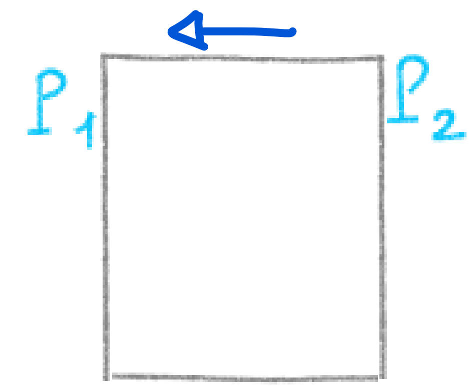
1) ELEMENTO INFINITESIMO
2) SEZIONE LONGITUDINALE
3) EQUILIBRIO ALLA TRASLAZIONE IN Z (PARTE CAMPITA *)

RISULTANTE SFORZI NORMALI + RISULTANTE SFORZI TANGENZIALI = 0

$$N^* + dN^* - N^* + dH = 0$$

$$\hookrightarrow dN^* + dH = 0$$





SOSTITUENDO

$$\frac{T_y}{I_x} S_x^* = - \tau_m s(t)$$

$$\boxed{\tau_m = - \frac{T_y}{s(t) I_x} S_x^*}$$

FORMULA DI JOURAWSKY !!!

TAGLIO RETTO: TENSIONI TANGENZIALI ALLA JOURAWSKY

EQUILIBRIO ALLA JOURAWSKY $\Longrightarrow$ CIOCCOLATINO PERUGINA

1) ELEMENTO INFINITESIMO
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RISULTANTE SFORZI NORMALI + RISULTANTE SFORZI TANGENZIALI = 0

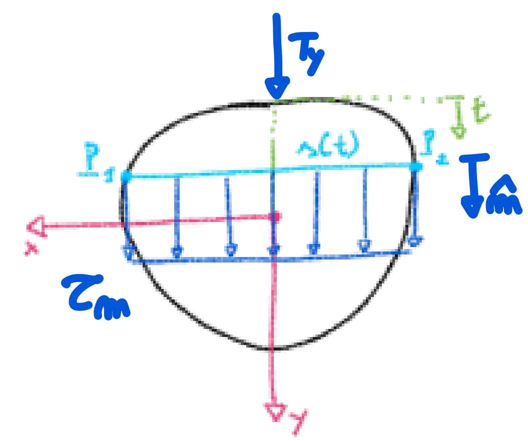
$$\boxed{\boxed{\tau_m = -\frac{T_y}{s(t)\, I_x} S_x^*}}$$

FORMULA DI JOURAWSKY PER TAGLIO RETTO SECONDO y

$$\boxed{\tau_m(t) = -\frac{T_x}{s(t)\, I_y} S_y^*}$$

F. DI J. PER T.R. SECONDO x

- $T_y > 0$ se concorde con y
- S_x^* DELLA PARTE SUPERIORE ($S_x^{**} = -S_x^*$)



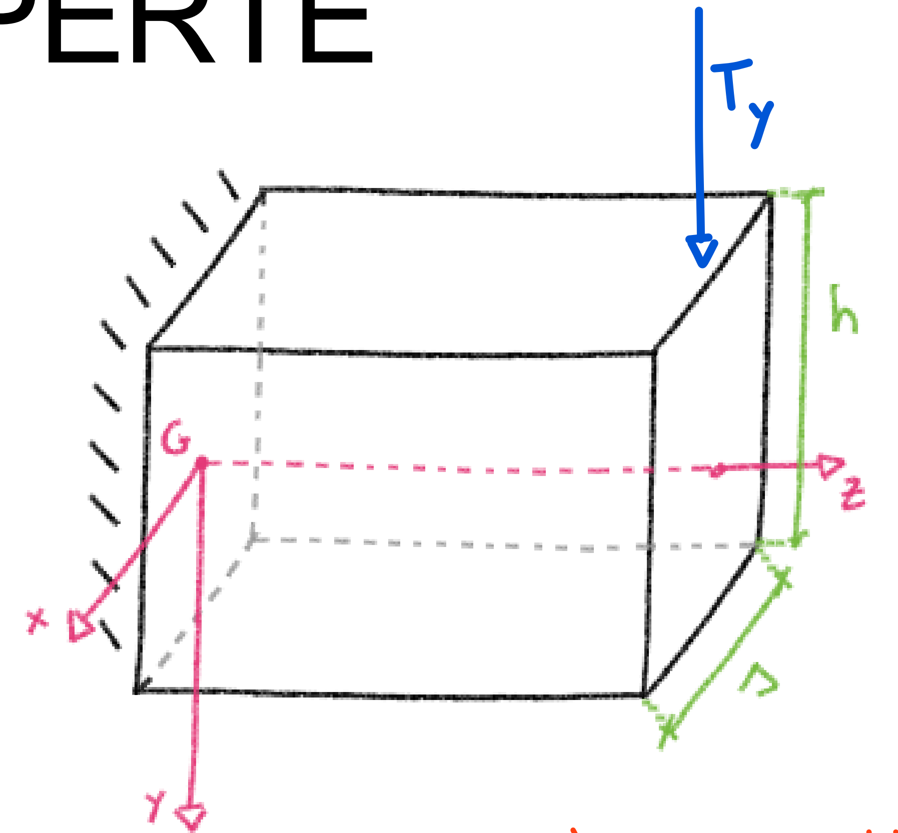
⊙ VALIDA SE $\tau_{zx} \sim 0$ E τ_m SE PICCOLO SPESSORE

⊙ TEORIA APPROSSIMATA

SEZIONI SOTTILI APERTE

Sezione Rettangolare Sottile : $s << h$

(Studio Flessione e Taglio)



$$\underline{\underline{T}} = \begin{bmatrix} 0 & 0 & \tau_{xz} \\ 0 & 0 & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} ; \quad \sigma_z = \frac{M_x}{I_x} y = -\frac{T_y(l-z)}{I_x} y$$

$$\left(I_x = \frac{1}{12} s h^3 \right)$$

Hp. Jourawsky: $\tau_{zx} = 0$; $\tau_m(t) = -\frac{T_y}{I_x} \frac{S_x^*}{s(t)}$

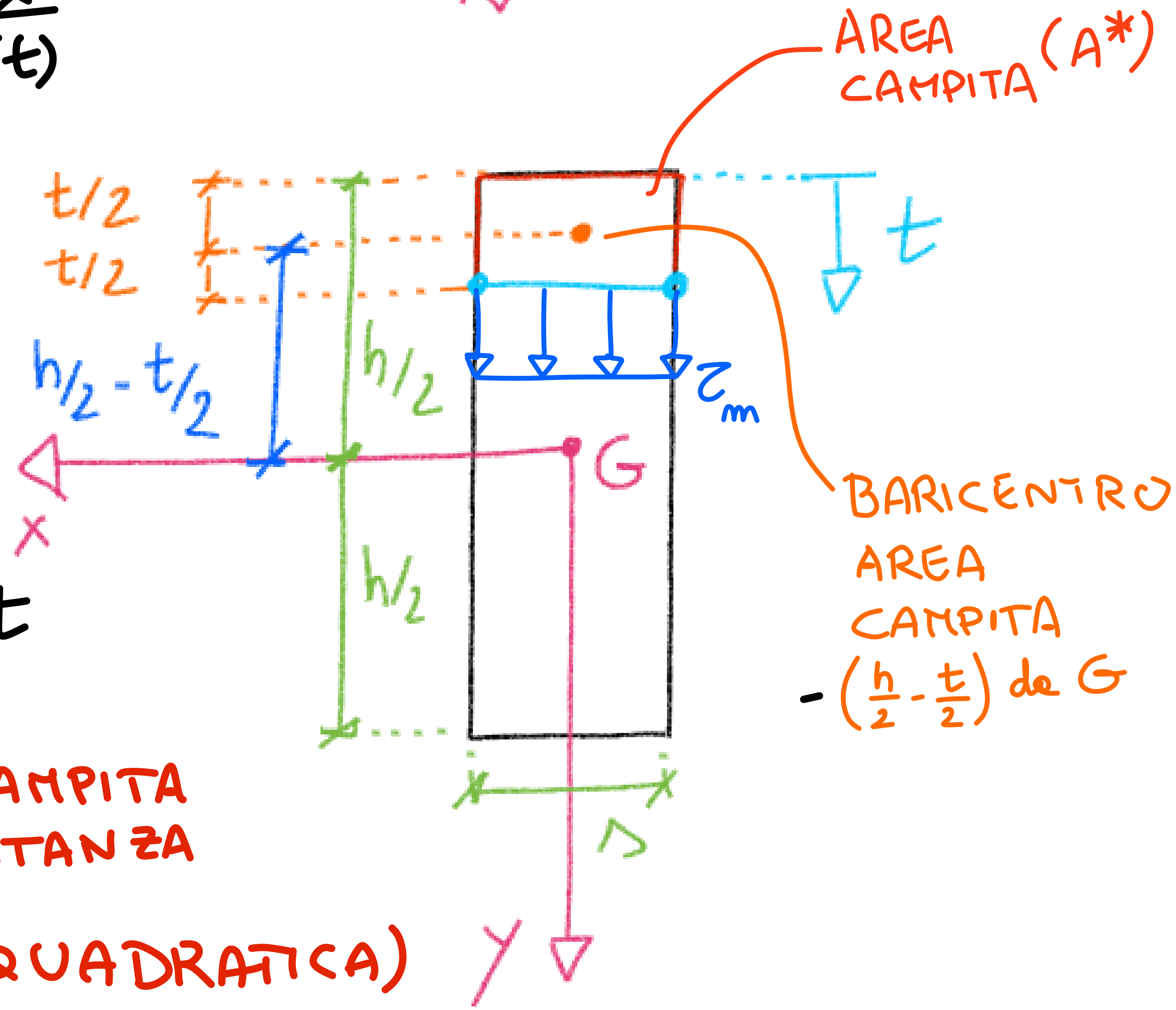
↳ **Calcolo S_x^* :**

$$S_x^*(t) = \int_{A^*} y \, dA^* = \int_{A^*} -\left(\frac{h}{2} - \frac{t}{2}\right) dA^*$$

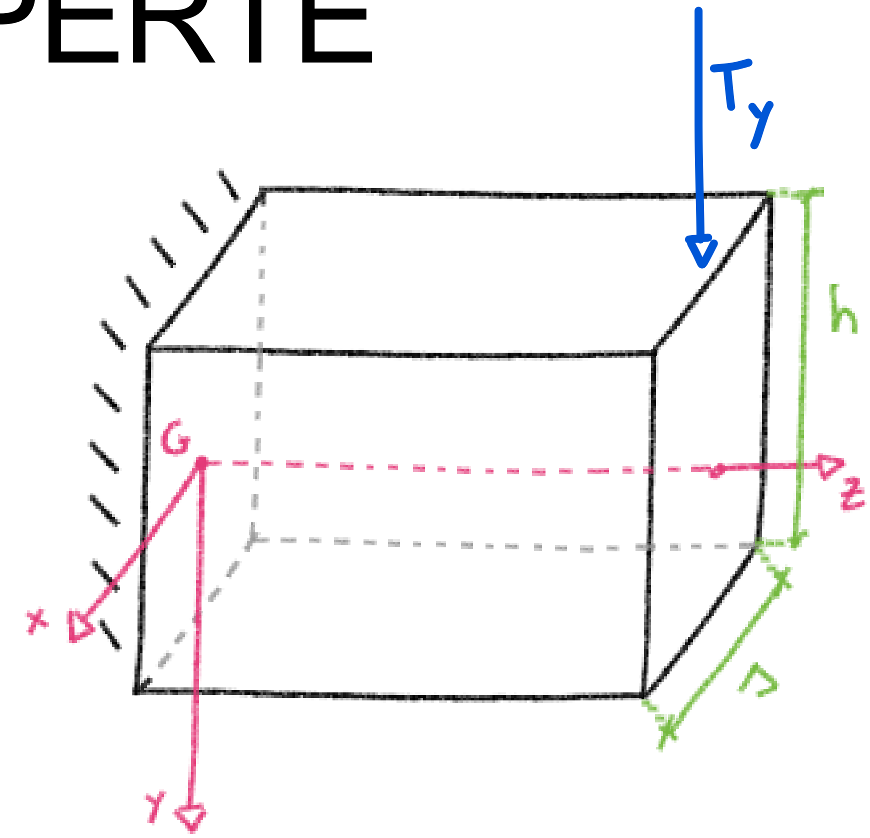
↳ $$S_x^*(t) = -\left(\frac{h}{2} - \frac{t}{2}\right) A^* = -\left(\frac{h}{2} - \frac{t}{2}\right) s t$$

↳ $$\boxed{S_x^*(t) = -\frac{s}{2}(h-t)\,t}$$ = AREA CAMPITA × DISTANZA

(FUNZ. QUADRATICA)



SEZIONI SOTTILI APERTE



SEZIONE RETTANGOLARE SOTTILE : $s \ll h$

(STUDIO FLESSIONE E TAGLIO)

$$\underline{\underline{T}} = \begin{bmatrix} 0 & 0 & \tau_{xz} \\ 0 & 0 & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} ; \quad \sigma_z = \frac{M_x}{I_x} y = -\frac{T_y (l-z)}{I_x} y$$

$$\left(I_x = \frac{1}{12} s h^3 \right)$$

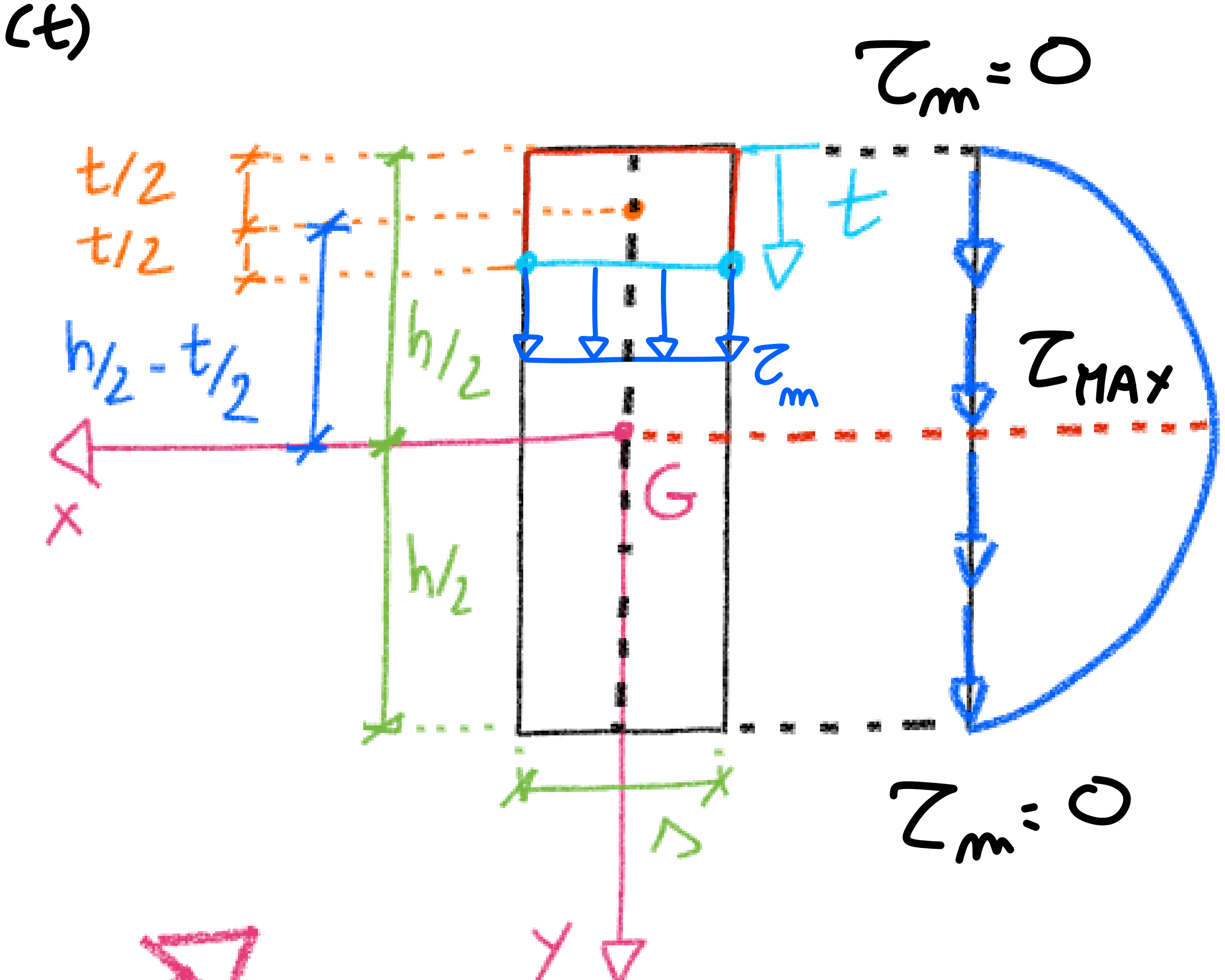
Hp. JOURAWSKY: $\tau_{zx} = 0$; $\tau_m(t) = -\frac{T_y}{I_x} \frac{S_x^*}{s(t)}$

$$\boxed{S_x^*(t) = -\frac{s}{2}(h-t)\,t}$$

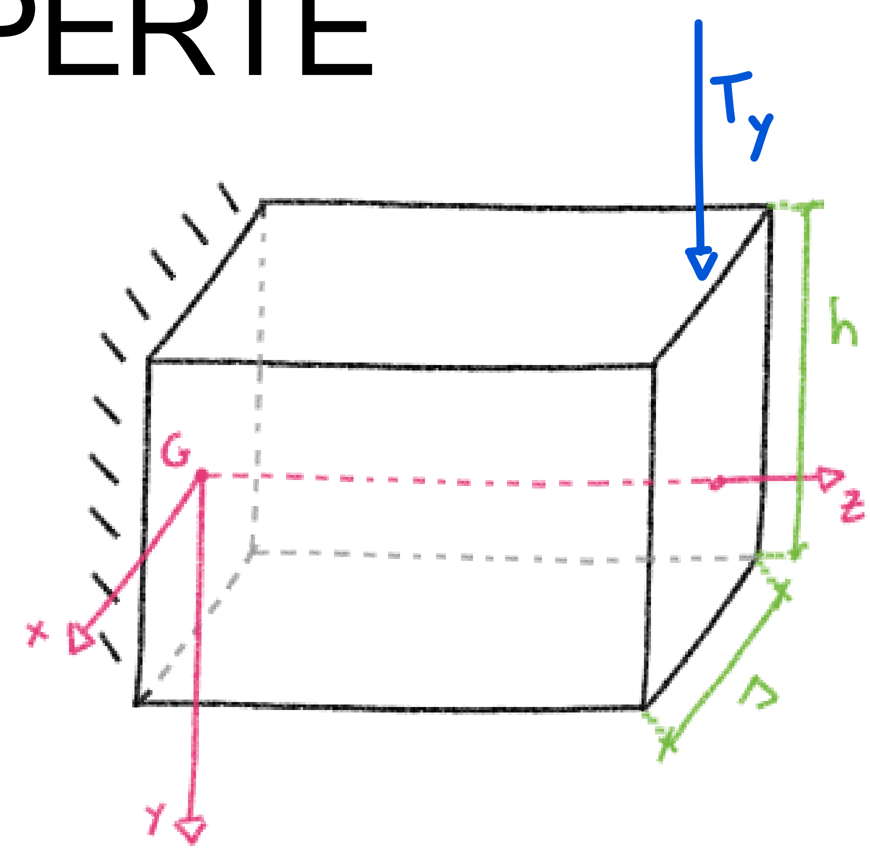
$\Downarrow$

$$\tau_m(t) = \frac{6 T_y}{s h^3} (h-t)\,t$$

DIAGRAMMA LUNGO t



SEZIONI SOTTILI APERTE

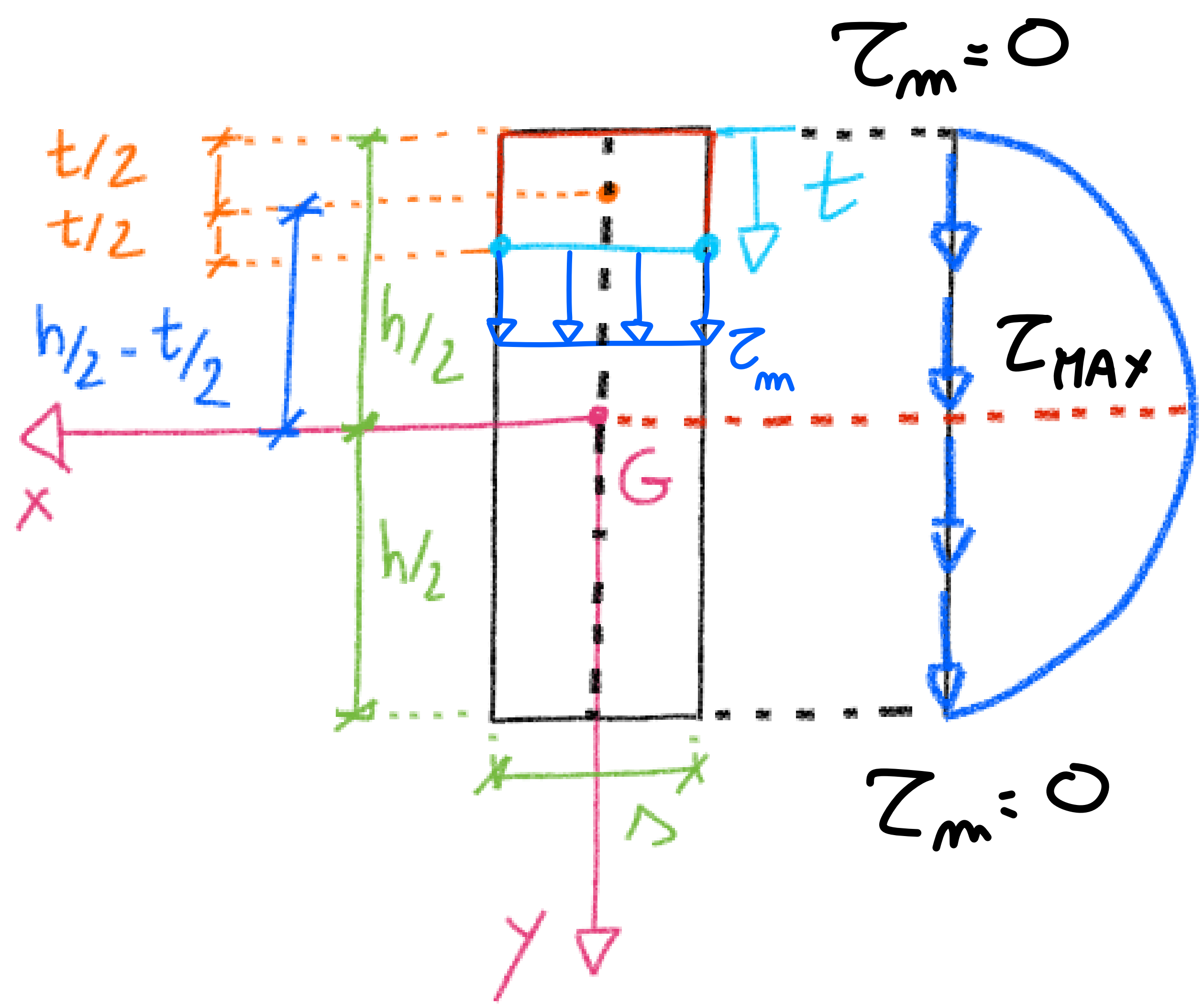


Sezione Rettangolare Sottile : $s \ll h$

(Studio Flessione e Taglio)

$$\underline{\underline{T}} = \begin{bmatrix} 0 & 0 & \tau_{xz} \\ 0 & 0 & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} ; \quad \sigma_z = \frac{M_x}{I_x} y = -\frac{T_y (l-z)}{I_x} y$$

$$\tau_m(t) = \frac{6 T_y}{s h^3} (h-t)\, t$$



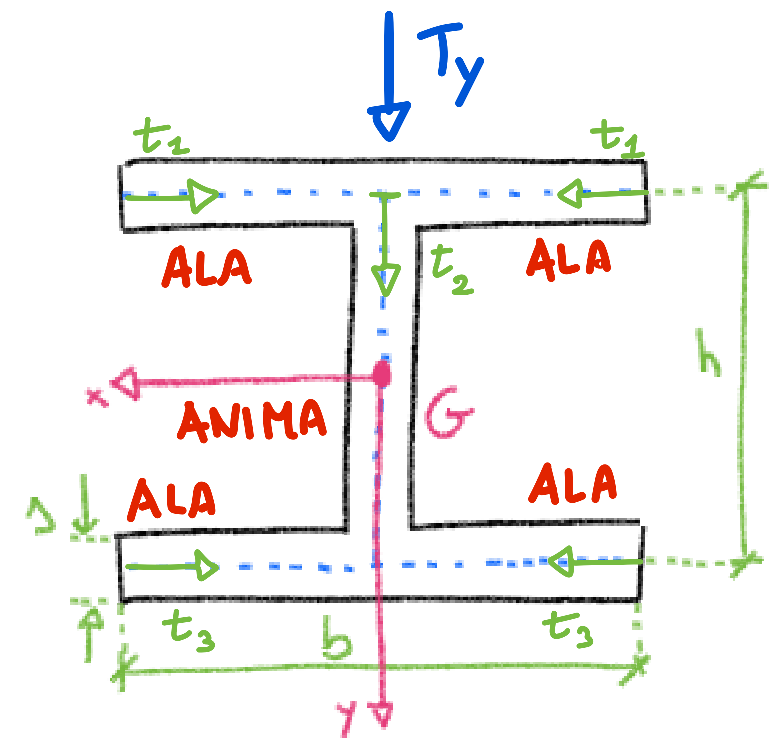
NOTE :

- $\tau_{MAX} \equiv \tau_m(t|_G)$
- $\tau_m = 0$ per $t|_{\text{ESTREMI LIBERI}}$
- DISTRIBUZIONE SIMMETRICA

SEZIONI SOTTILI APERTE

Sezione a Doppio T

$A = (2b+h)s \; ; \quad I_x = \left(\frac{bh^2}{2} + \frac{h^3}{12} \right) s$



(*) Divisione in base alla linea media

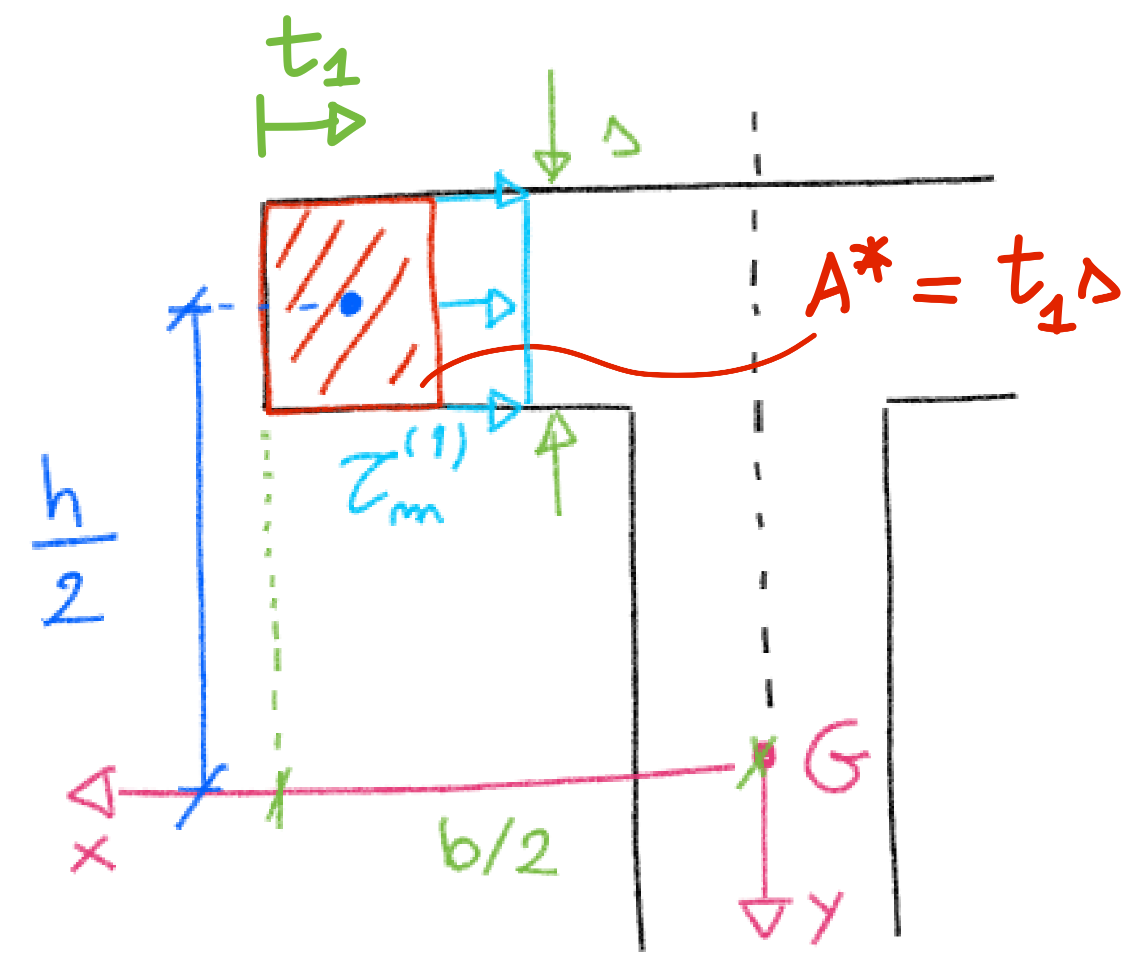
(*) Simmetria

1s) Ala Superiore : $0 \leq t_1 \leq b/2$

$S_x^{*(1)} = -\left(\frac{h}{2}\right) s\, t_1$

$\boxed{\text{FUNZIONE LINEARE}}$

$\tau_m^{(1)}(t_1) = -\frac{T_y}{I_x} \frac{S_x^{*(1)}}{s}$



$\tau_1 = \tau_m^{(1)}(t = b/2) = \tau_{max}$

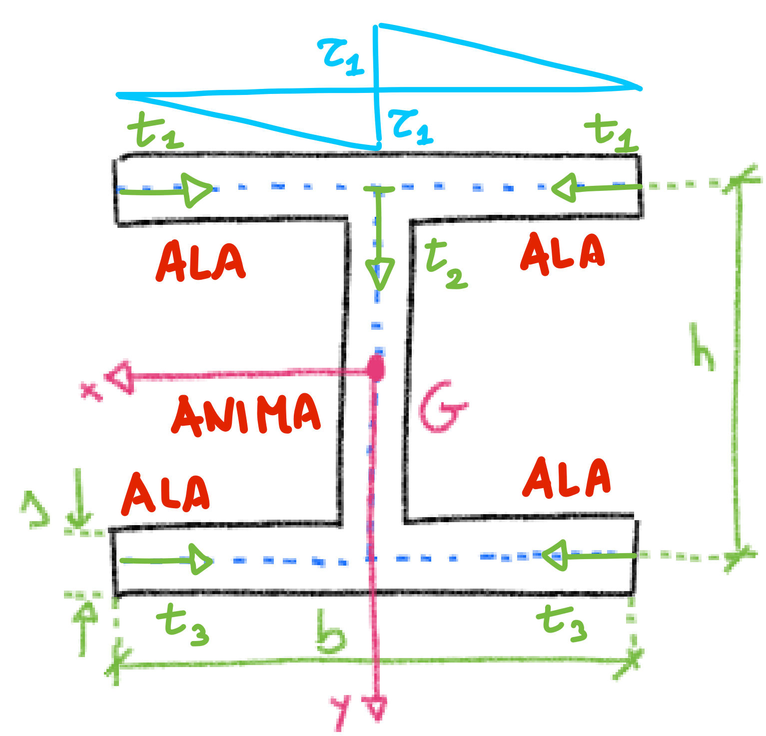
SEZIONI SOTTILI APERTE

Sezione a Doppio T

$A = (2b+h)s \; ; \quad I_x = \left(\frac{b h^2}{2} + \frac{h^3}{12} \right) s$

(*) Divisione in base alla linea media

(*) Simmetria



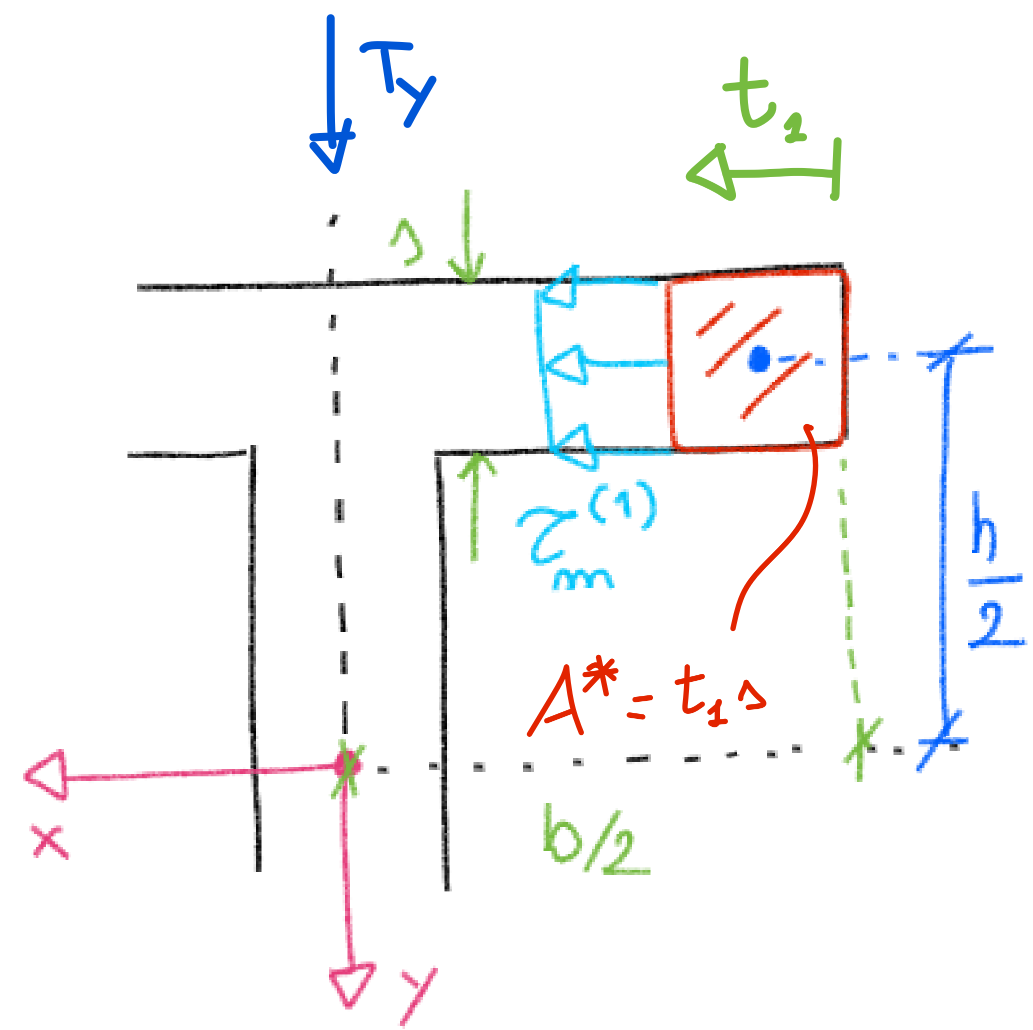
1^s) Ala Superiore : $0 \le t_1 \le b/2$

$$S_x^{*(1)} = -\left(\frac{h}{2}\right) s\, t_1$$

$$\tau_m^{(1)}(t_1) = -\frac{T_y}{I_x}\,\frac{S_x^{*(1)}}{s}$$

$\boxed{\text{FUNZIONE LINEARE}}$

$$\tau_1 = \tau_m^{(1)}(t = b/2) = \tau_{max}$$



1^D) Ala Superiore : $0 \le t_1 \le b/2$

$\Rightarrow$ simmetrica (segno opposto)

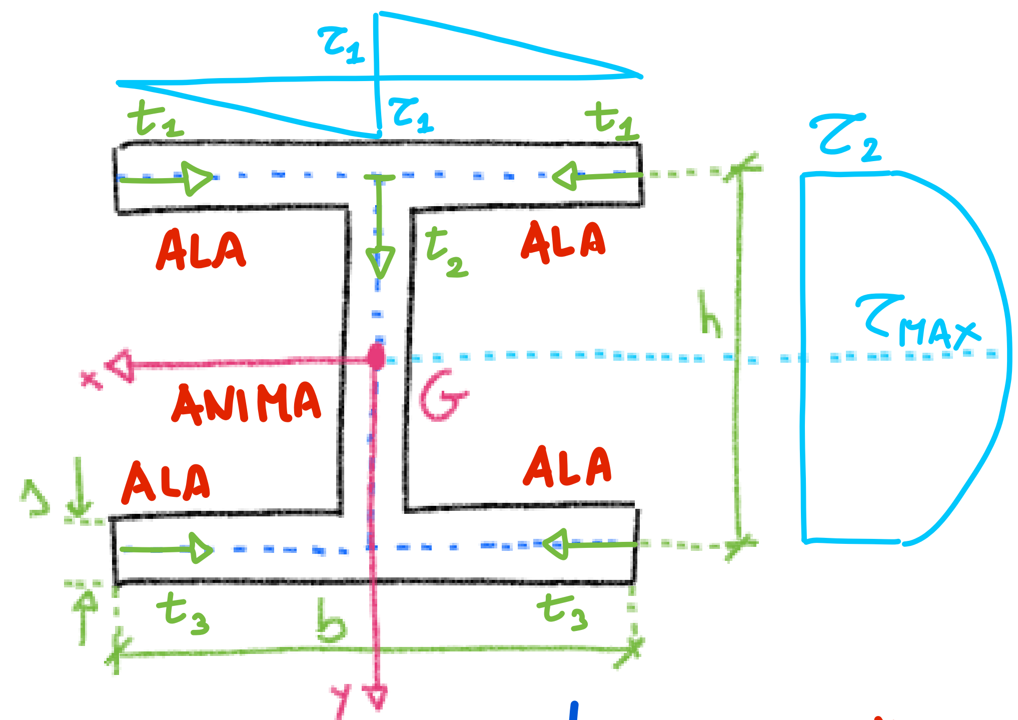
SEZIONI SOTTILI APERTE

SEZIONE A DOPPIO T

$A = (2b+h)s$; $I_x = \left(\frac{bh^2}{2} + \frac{h^3}{12}\right)s$

(*) DIVISIONE IN BASE ALLA LINEA MEDIA

(*) SIMMETRIA

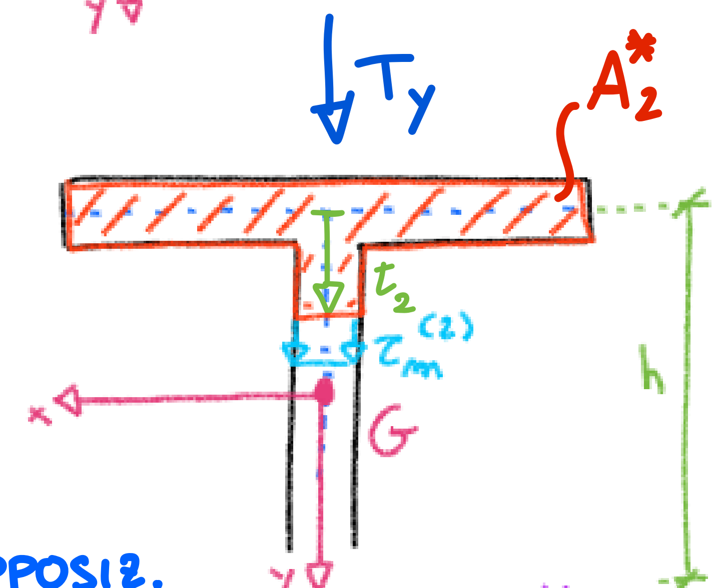


2) ANIMA: $0 \le t_2 \le h$

$$S_x^{*(2)} = -\frac{h}{2}\cdot sb - \left(\frac{h}{2} - \frac{t_2}{2}\right) s\, t_2$$

$$\tau_m(t_2) = -\frac{T_y}{I_x}\,\frac{S_x^{*(2)}}{s}$$

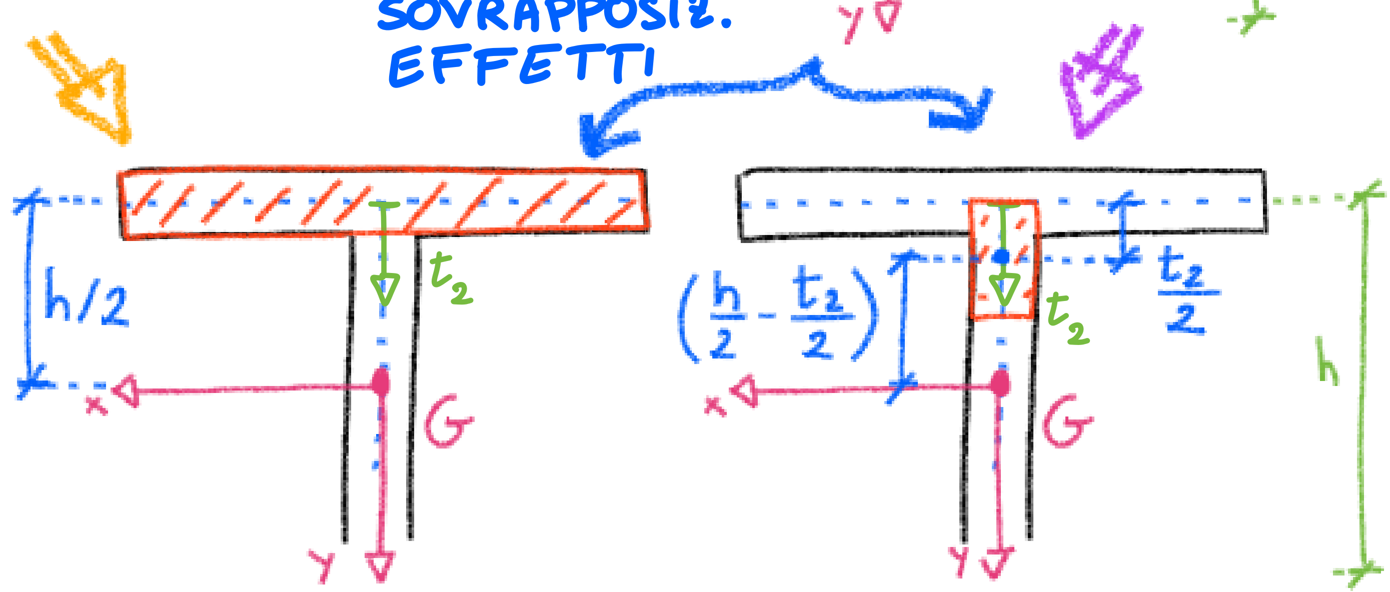
$\boxed{\text{FUNZIONE PARABOLICA}}$



$$\tau_2 = \tau_m(t_2 = 0) = 2\tau_1$$

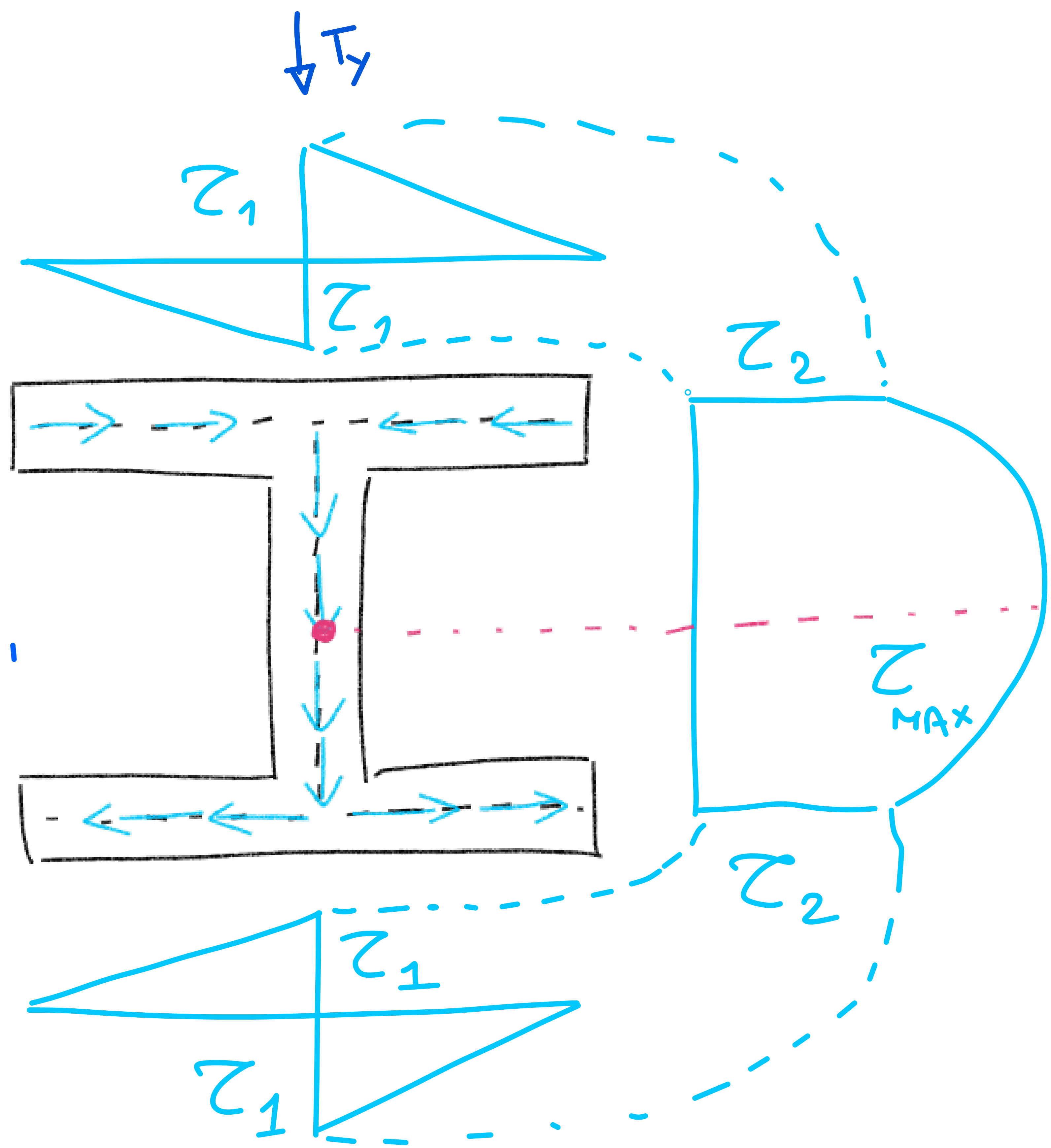
(se $s_1 \neq s_2$ ⇒ $\tau_2 = 2\tau_1 s_1/s_2$)

$$\tau_{MAX} = \tau_m(t_2|_G)$$



SEZIONI SOTTILI APERTE

SEZIONE A DOPPIO T



NOTE :

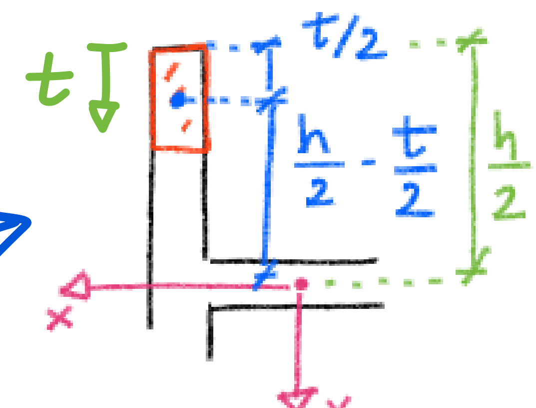
- $\tau_{MAX} \equiv \tau_m (t|_G)$
- $\tau_m = 0$ per $t|_{\text{ESTREMI LIBERI}}$
- DISTRIBUZIONE SIMMETRICA

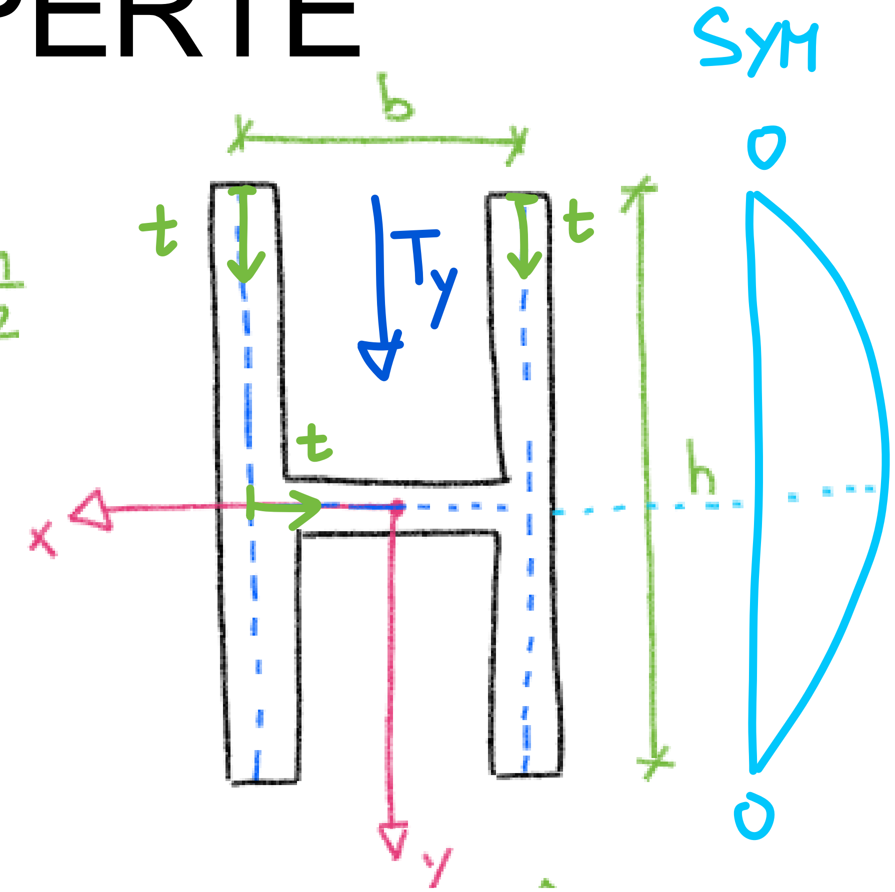
SEZIONI SOTTILI APERTE

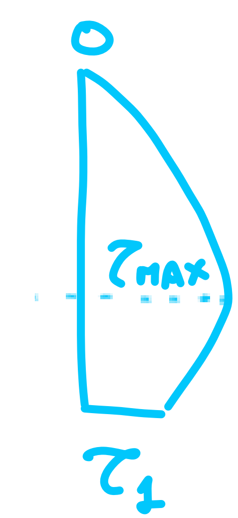
SEZIONE AD U, H : $I_x = \dfrac{s h^3}{6}$

$$S_x^*(t) = -\left(\frac{h}{2} - \frac{t}{2}\right) s\, t \qquad (0 \le t \le h)$$

$\tau_m(t)$ PARABOLICO $\begin{cases} \tau_m(0) = 0 \\ \tau_m(G) = \tau_{max} \end{cases}$







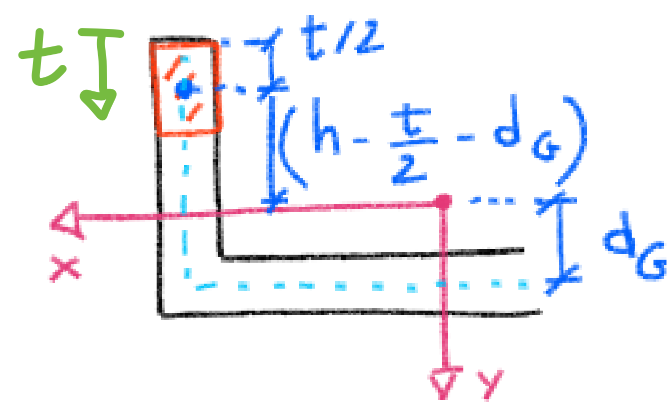
$$A^* = s\,t \;\rightarrow\; S_x^* = -\left(h - \frac{t}{2} - d_G\right) s\, t$$

$$\tau_m(t) = -\frac{T_y}{I_x}\frac{S_x^*}{s} = \frac{T_y}{I_x}\left(h - \frac{t}{2} - d_G\right) t$$

$$\tau_m(0) = 0$$

$$\tau_1 = \tau_m(t = h) = \frac{T_y}{I_x}\left(\frac{h}{2} - d_G\right) h \neq 0$$

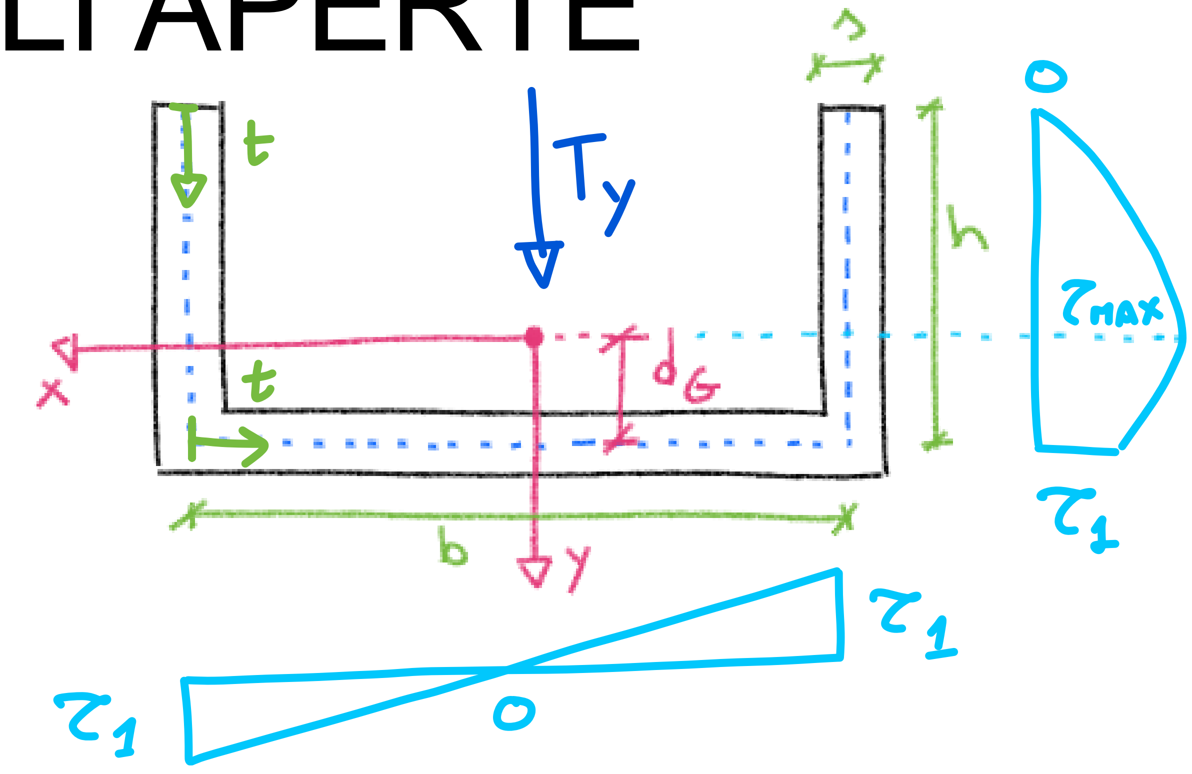
$$\tau_{MAX} = \tau_m(t = h - d_G) = \frac{T_y}{I_x}\left(\frac{h}{2} - \frac{d_G}{2}\right)(h - d_G)$$



SEZIONI SOTTILI APERTE

Sezione ad U, H:

$$d_G = \frac{h^2}{2h+b} \; ; \; I_x = \frac{h^3(h+2b)}{3(2h+b)}s$$



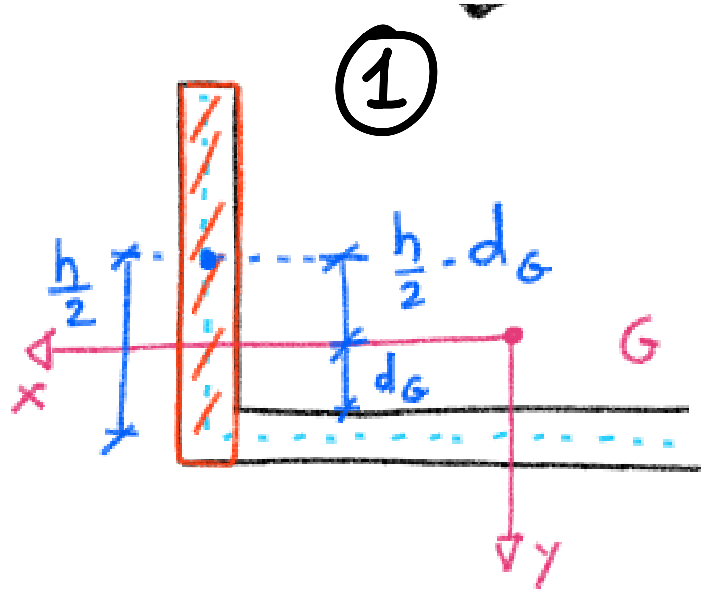
① $A_1^* = sh \rightarrow S_x^{*(1)} = -\left(\frac{h}{2} - d_G\right) sh$ = cost

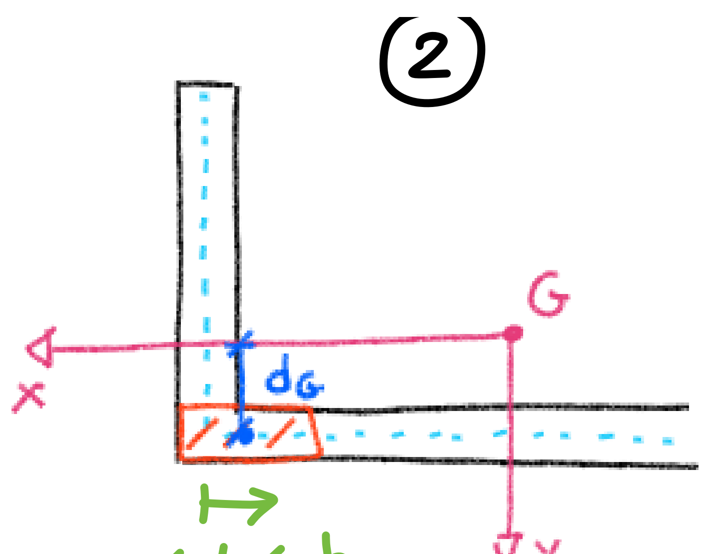
② $A_2^* = st \rightarrow S_x^{*(2)} = \int_{A^*} y \, dA^* = d_G s t$ = lin

$$S_x^* = S_x^{*(1)} + S_x^{*(2)} =$$

$$= -\left(\frac{h}{2} - d_G\right) sh + d_G s t$$

$$\tau_m(t) = \frac{T_y}{I_x}\left[-\left(\frac{h}{2} - d_G\right)h + d_G t\right]$$

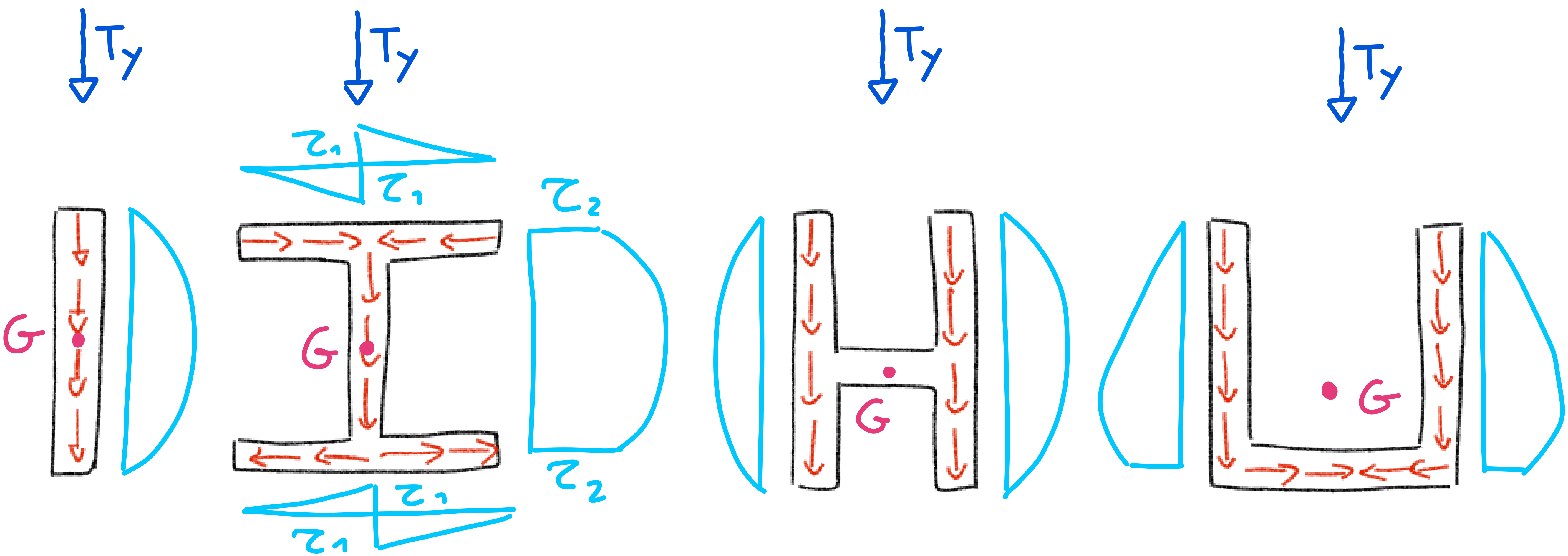




SEZIONI SOTTILI APERTE

CARATTERISTICHE

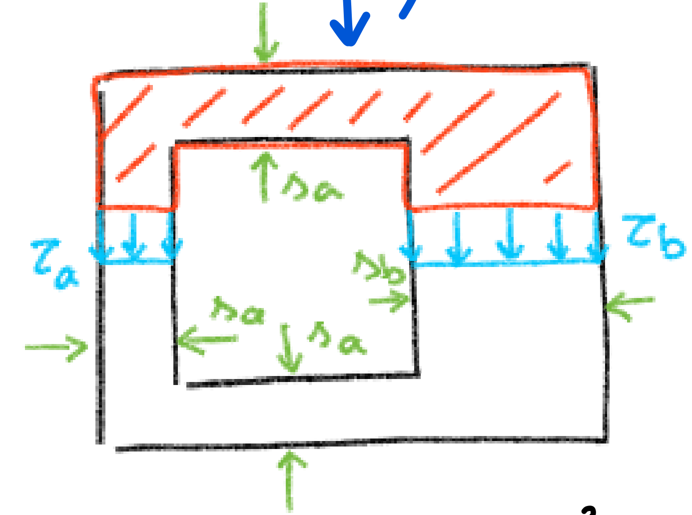
- τ // LINEA MEDIA e COSTANTI SULLO SPESSORE
- $q = s(t)\,\tau_m(t)$
- ESTREMI LIBERI : $\tau_m = 0$
- ASSE NEUTRO : $\tau_m = \tau_{MAX}$
- NO SOLENOIDALE



SEZIONI SOTTILI CHIUSE

(*) IN GENERALE E' NECESSARIO UTILIZZARE CONDIZIONI DI COMPATIBILITÀ CINEMATICA

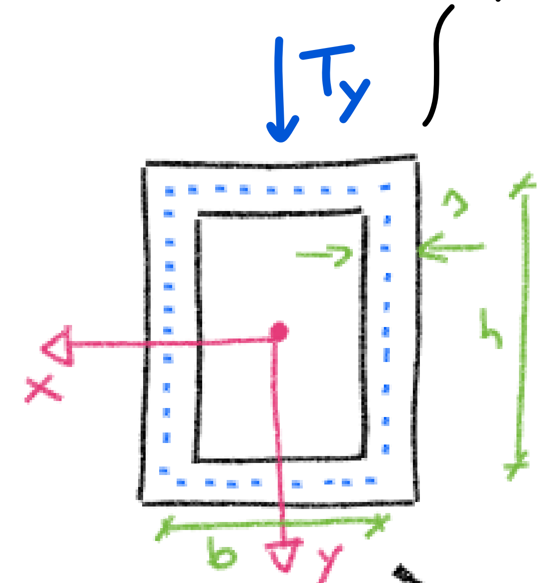
$$\tau_a s_a + \tau_b s_b = -\frac{T_y S_x^*}{I_x}$$



(*) SE SIMMETRICA E CARICATA SYM ALLORA E' SUFFICIENTE L'EQUIL.

$$I_x = 2\left(\frac{s h^3}{12} + s\frac{b h^2}{4}\right)$$

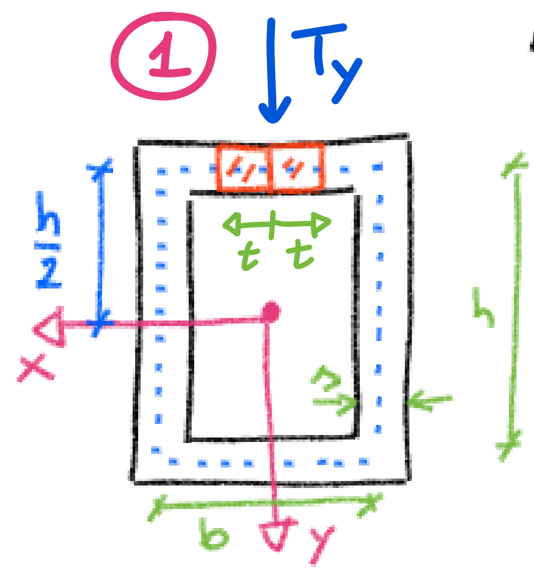
$$\tau_m(t) = -\frac{1}{2}\frac{T_y S_x^*}{I_x s} \qquad \begin{cases} s_a = s_b \\ \tau_a = \tau_b = \tau_m(t) \end{cases}$$

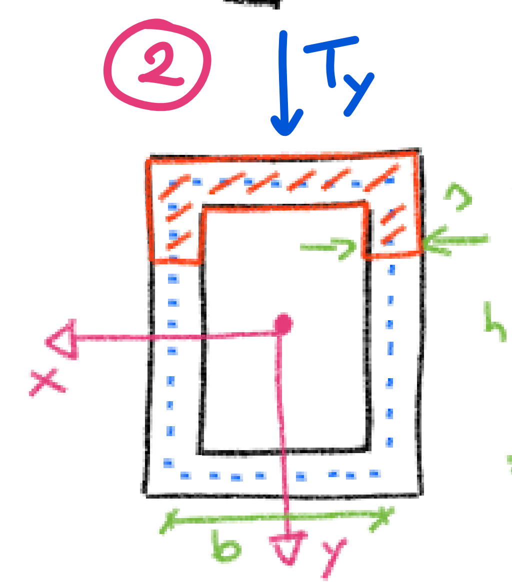


(1) $0 \le t \le b/2$

$$S_x^{*(1)} = -\frac{h}{2}(s t)\cdot 2 = -s h t$$

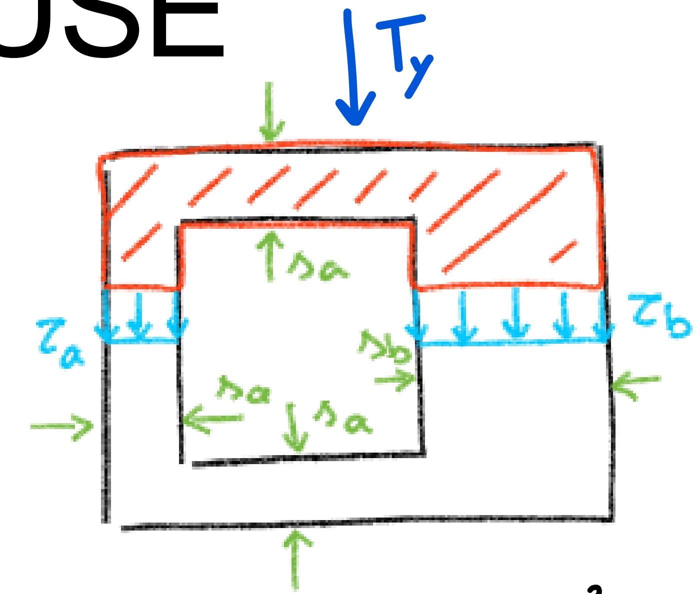
$$\tau_m^{(1)}(t) = \frac{T_y}{I_x}\frac{h t}{2}$$





(2) → DUE CONTRIBUTI

SEZIONI SOTTILI CHIUSE

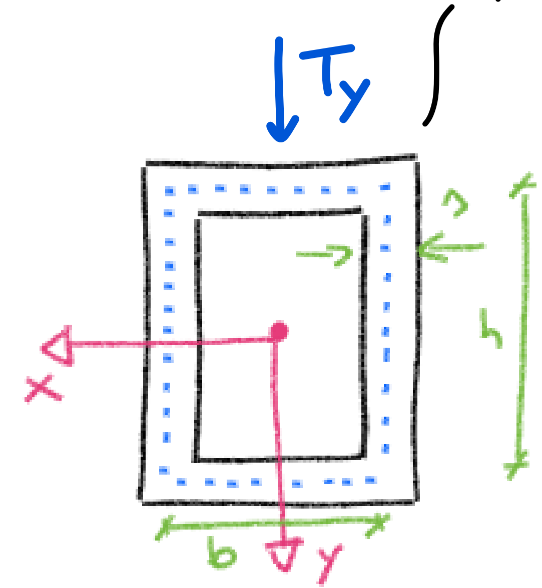


$$\tau_a s_a + \tau_b s_b = -\frac{T_y S_x^*}{I_x}$$

(*) IN GENERALE E' NECESSARIO UTILIZZARE CONDIZIONI DI COMPATIBILITÀ CINEMATICA

(*) SE SIMMETRICA E CARICATA SYM ALLORA E' SUFFICIENTE L'EQUIL.

$$I_x = 2\left(\frac{s h^3}{12} + \frac{s b h^2}{4}\right)$$

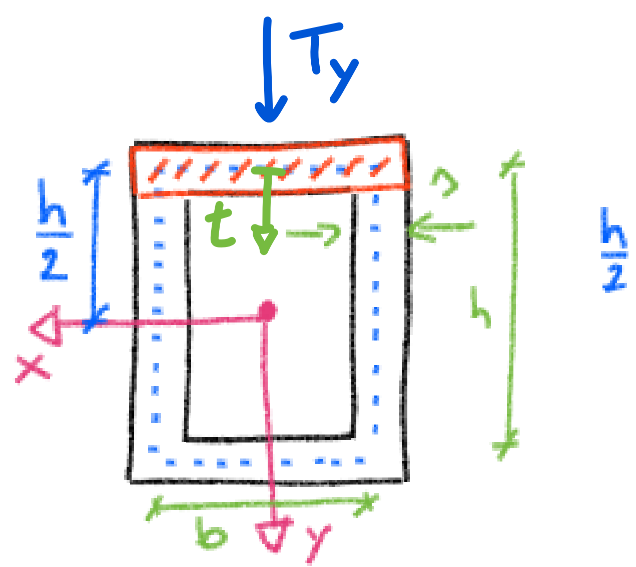


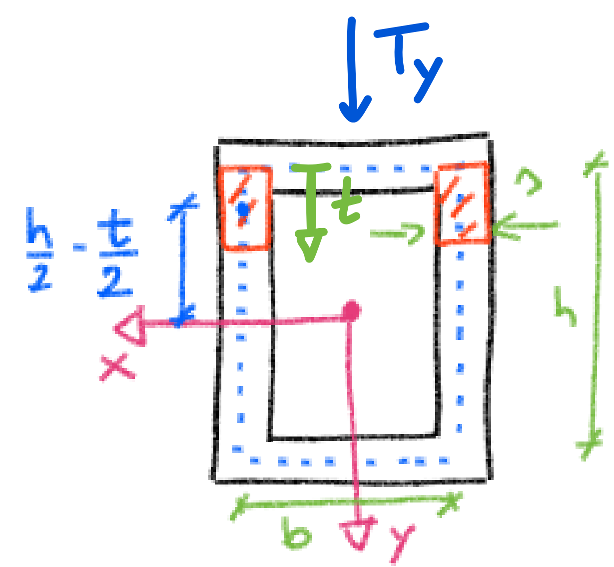
$$\tau_m(t) = -\frac{1}{2}\frac{T_y S_x^*}{I_x s}$$

$$\begin{cases} s_a = s_b \\ \tau_a = \tau_b = \tau_m(t) \end{cases}$$

$$0 \leq t \leq h$$

$$\left.\begin{array}{l} 2^1 \quad S_x^{*(2^1)} = -\frac{h}{2} s b \\ 2^2 \quad S_x^{*(2^2)} = -\left(\frac{h}{2} - \frac{t}{2}\right) s t \cdot 2 \end{array}\right\} S_x^{*(2)} = \Sigma_{1,2}$$





$$\tau_m^{(2)}(t) = \frac{T_y}{4 I_x}\left[bh + 2t(h-t)\right]$$

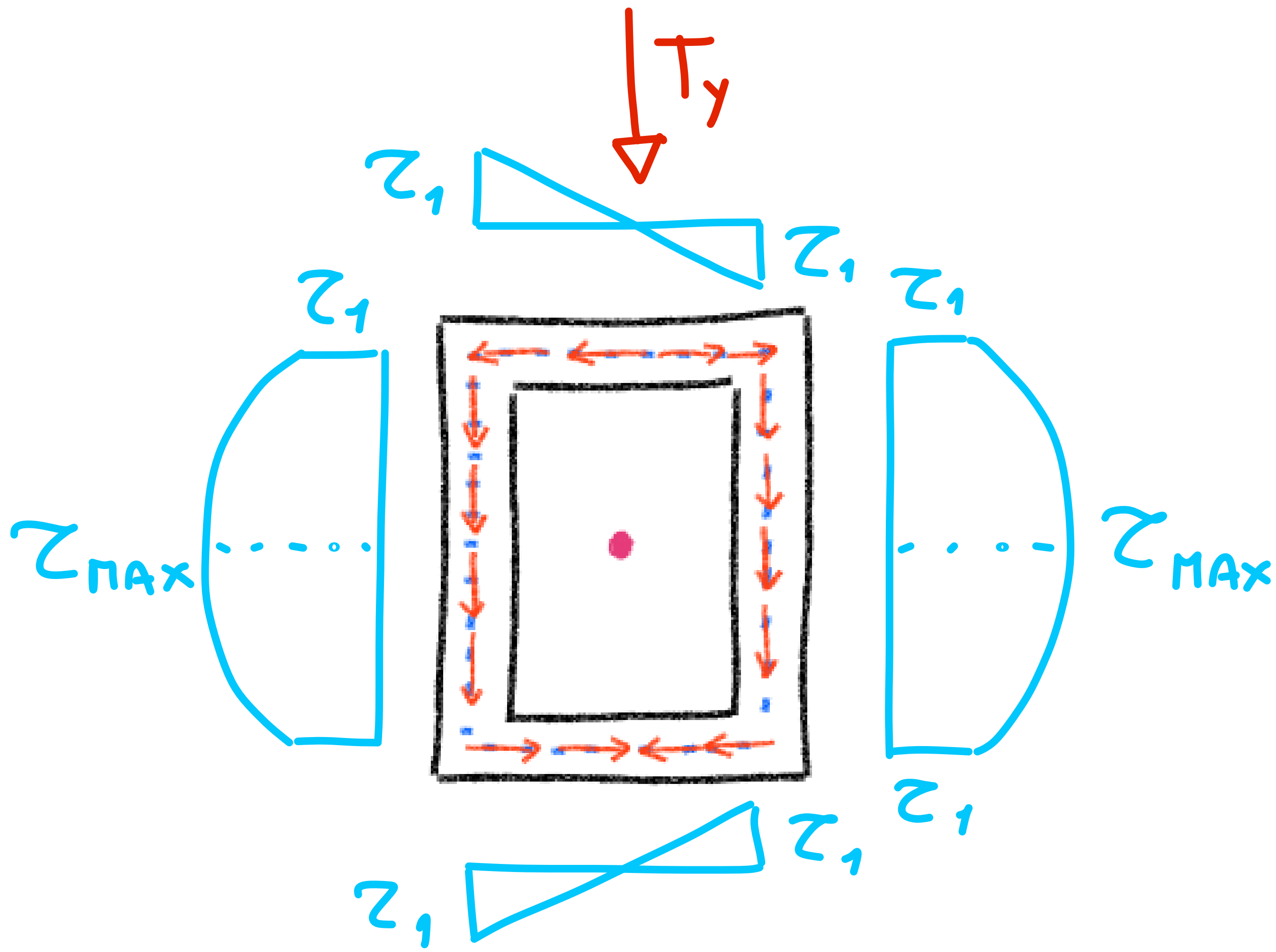
SEZIONI SOTTILI CHIUSE

$$\tau_m^{(1)}(t) = \frac{T_y}{I_x}\frac{h}{2}t$$

$$\tau_1 = \tau_m^{(1)}\left(\frac{b}{2}\right) = \frac{T_y}{2I_x}\frac{hb}{2}$$

$$\tau_m^{(2)}(t) = \frac{T_y}{4I_x}\left[bh + 2t(h-t)\right]$$

$$\tau_2 = \tau_{MAX} = \tau_m^{(2)}\left(t = \frac{h}{2}\right)$$



TAGLIO DEVIATO

T_y NON // ASSE PRINCIPALE D'INERZIA

$$\underline{T} = T_x \hat{i} + T_y \hat{j}$$

$\downarrow$ P.S.E.

$$\tau_m(t) = -\frac{T_y S_x^*(t)}{I_x s(t)} - \frac{T_x S_y^*(t)}{I_y s(t)}$$

$+$

$$\sigma_z = \frac{M_x}{I_x} y - \frac{M_y}{I_y} x \quad \Big\} \text{ FLESSIONE DEVIATA}$$

$\Rightarrow$ ATTIVAZIONE DI TUTTE LE COMPONENTI DI SFORZO DI dSV

SOLLECITAZIONE COMPOSTA

TAGLIO RETTO e TORSIONE

- SE y NON E' ASSE DI SIMMETRIA ALLORA T_y INDUCE FLESSIONE - TAGLIO e TORSIONE (APPROX. PER PICCOLO SPESSORE)

CENTRO DI TAGLIO

$C_T(x_T, y_T)$

PUNTO DEL PIANO, C_T, TALE CHE SE T_y PASSA PER C_T ALLORA LA SEZIONE NON RUOTA

→ DIPENDE SOLO DA GEOMETRIA:

- $C_T \in$ ASSE DI SIMMETRIA
- $C_T \equiv G \iff$ 2 ASSI DI SIMMETRIA
- $C_T \equiv$ INTERSEZIONE RETTANGOLI SOTTILI CONVERGENTI

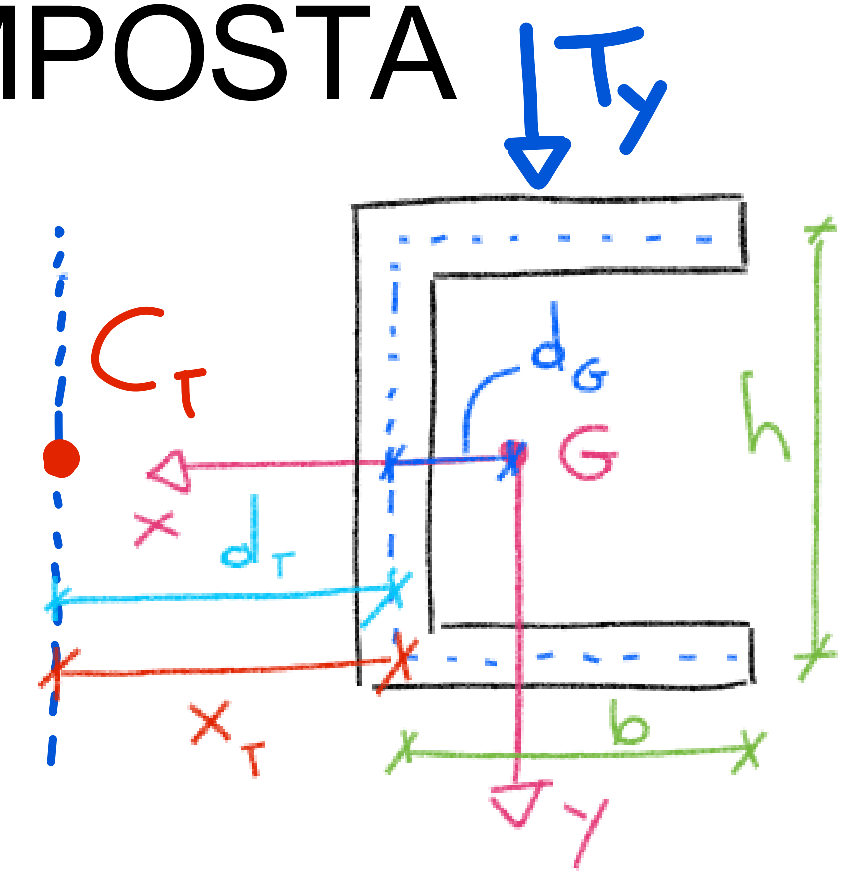
$\Rightarrow$ CASO SEZIONE SOTTILE C

SOLLECITAZIONE COMPOSTA

CENTRO DI TAGLIO

$C_T(x_T, y_T)$

PUNTO DEL PIANO, C_T, TALE CHE SE T_y PASSA PER C_T ALLORA LA SEZIONE NON RUOTA



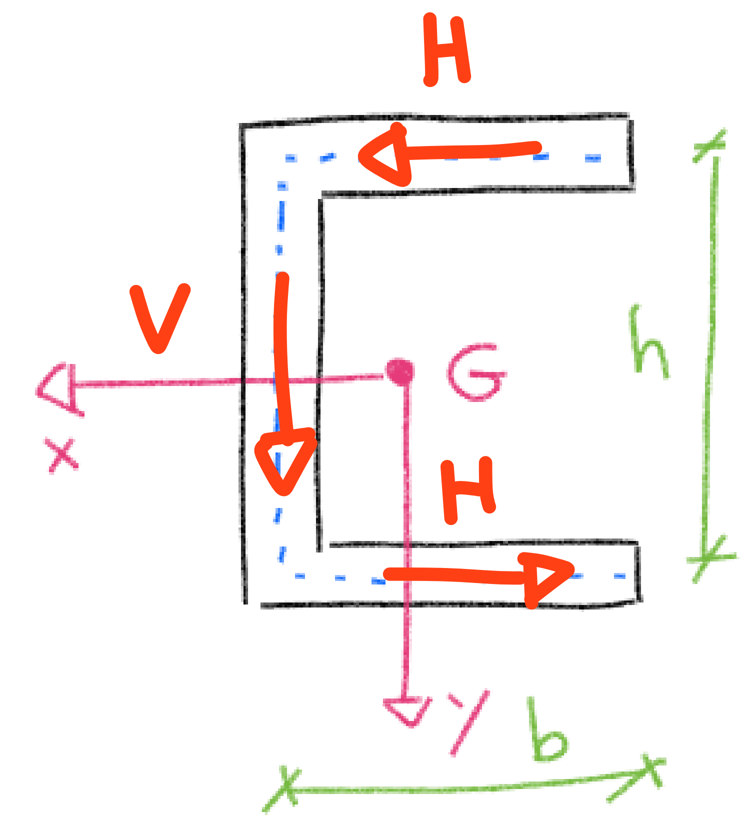
$$d_G = \frac{b^2}{2b+h} \; ; \quad I_x = \frac{s h^3}{12} + \frac{s b h^2}{2}$$

(*) C_T DEVE APPARTENERE AD ASSE x (SYM)

(*) CONSIDERO LE RISULTANTI DOVUTE AL SOLO TAGLI (τ):

H : risultante sulle ali

V : " sull'anima



↳ $V = T_y$ EQUIVALENZA STATICA

↳ $H = \int_0^b \tau^{(T_y)} s \, dt$

STUDIO

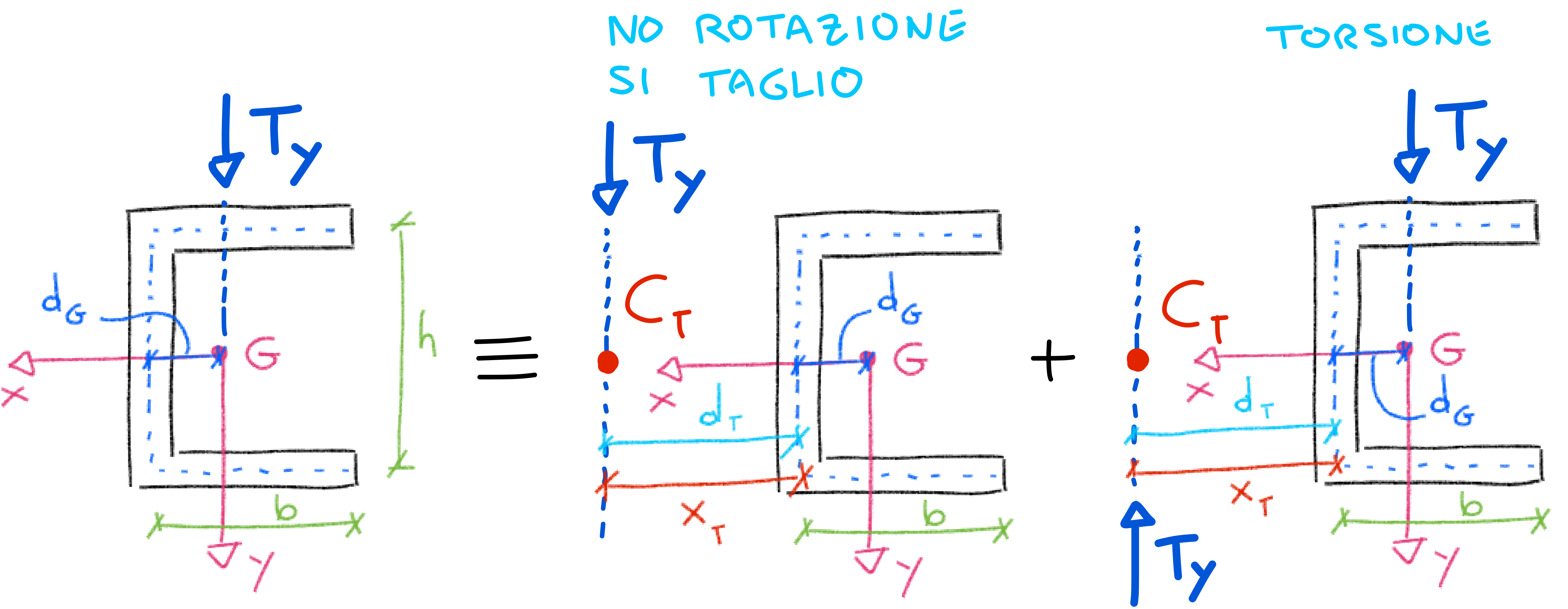
SOLLECITAZIONE COMPOSTA

TAGLIO RETTO e TORSIONE

- SE y NON E' ASSE DI SIMMETRIA ALLORA T_y INDUCE FLESSIONE - TAGLIO e TORSIONE (APPROX. PER PICCOLO SPESSORE)

STUDIO DISTR. SFORZI (Hp d_G NOTO)

$$\boxed{M_t = T_y \, x_T}$$

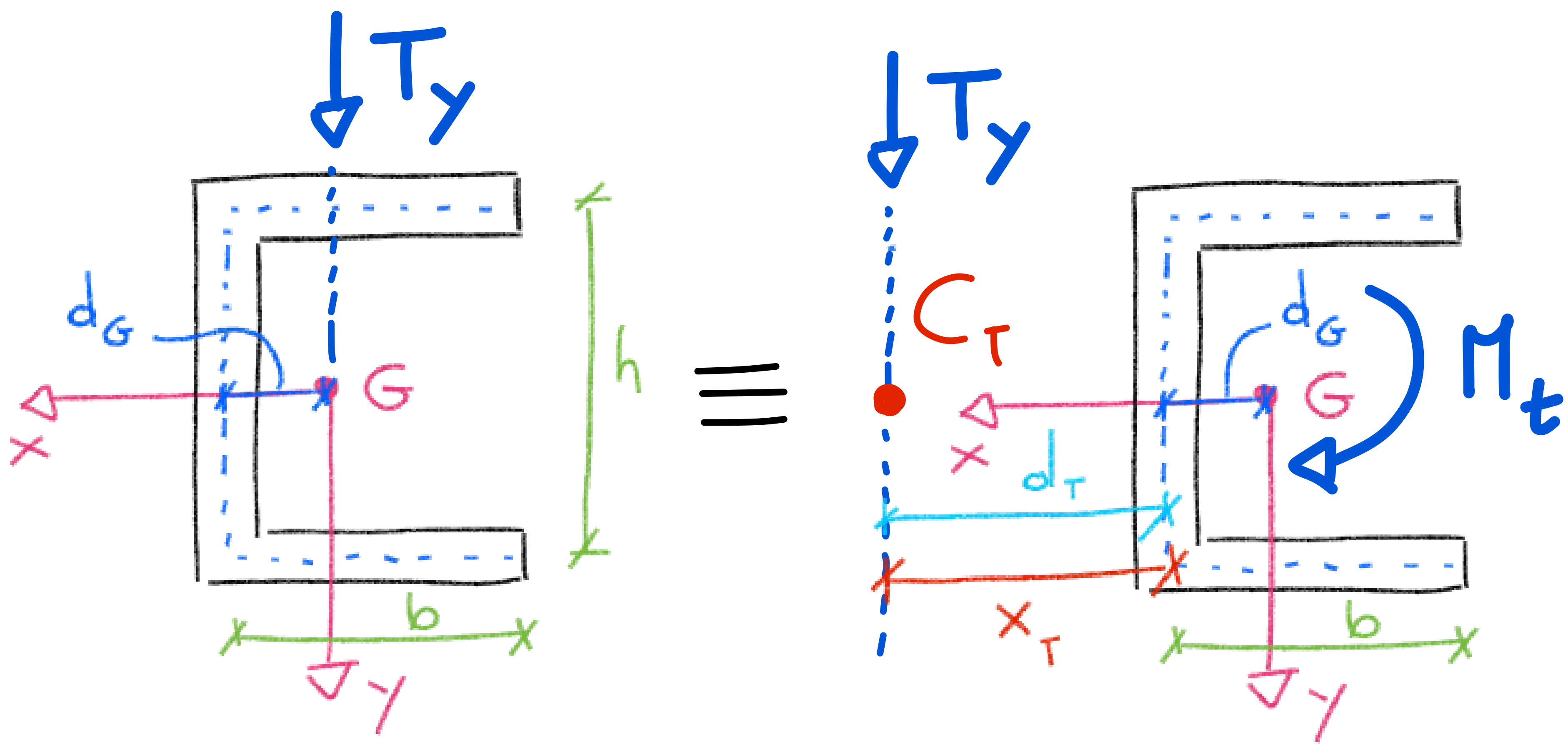


SOLLECITAZIONE COMPOSTA

STUDIO DISTR. SFORZI (Hp NOTO x_T)

$$\boxed{M_t = T_y \, x_T}$$

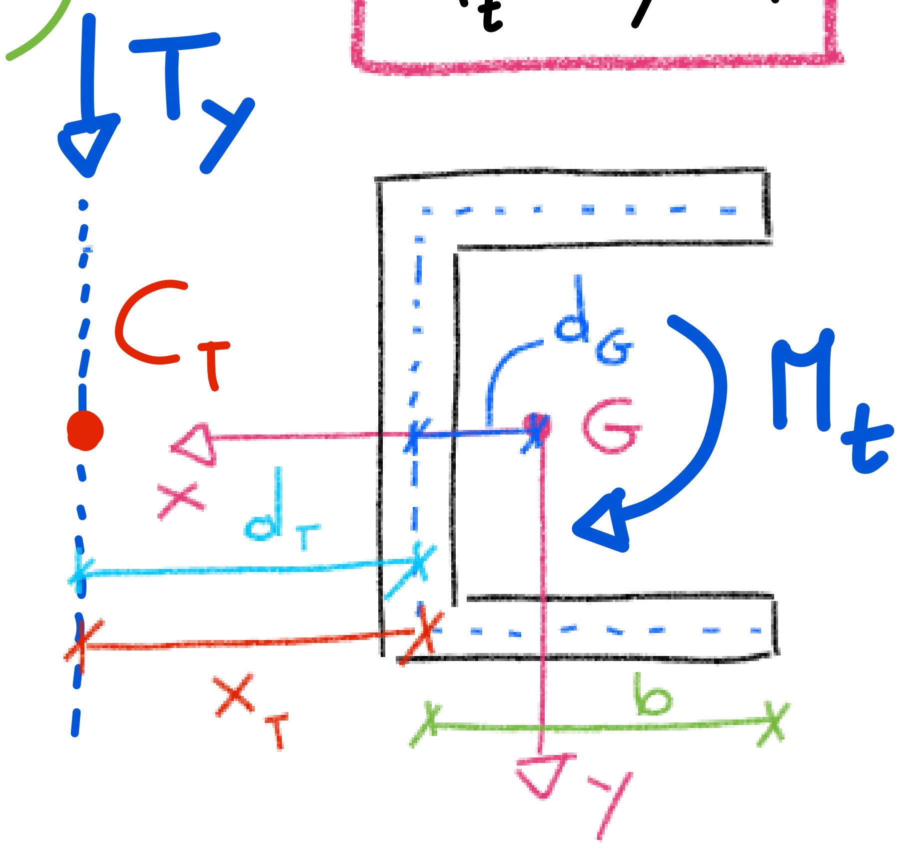
SOLLECITAZ. COMPOSTA $\equiv$ TAGLIO RETTO + TORSIONE



SOLLECITAZIONE COMPOSTA

STUDIO DISTR. SFORZI (Hp d_G NOTO)

$$\boxed{M_t = T_y \, x_T}$$



$$d_G = \frac{b^2}{2b+h} \; ; \quad I_x = \frac{s h^3}{12} + \frac{s b h^2}{2}$$

STUDIO TAGLIO :

1) $0 \le t \le b$: $S_x^* = -\frac{h}{2} s t$

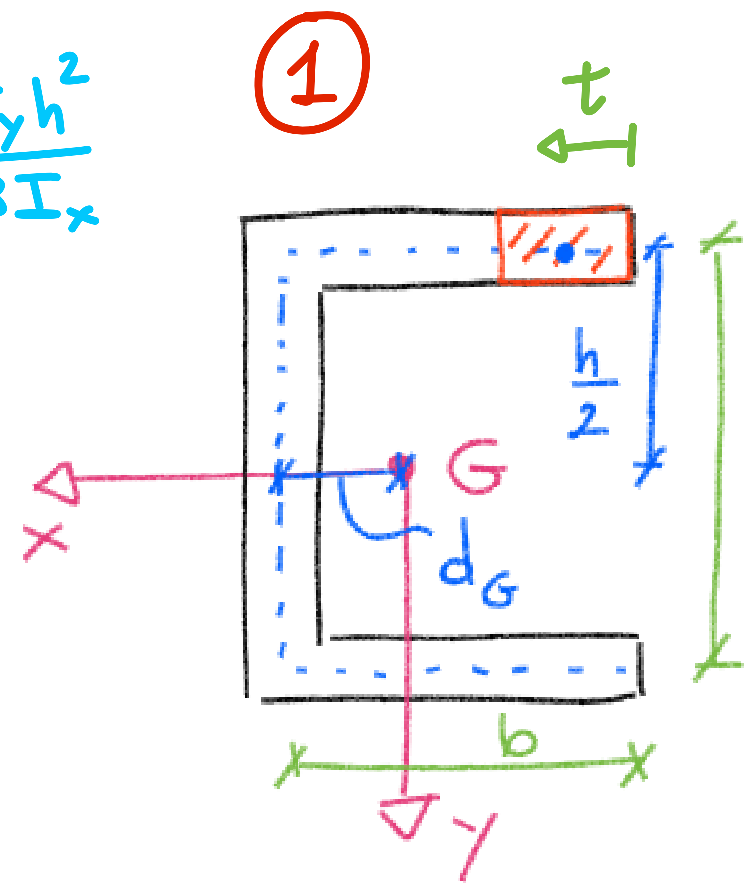
↳ $\tau_m^{(1)}(t) = -\frac{T_y S_x^*}{I_x s} = \frac{1}{2}\frac{T_y}{I_x} h t$ } LINEARE

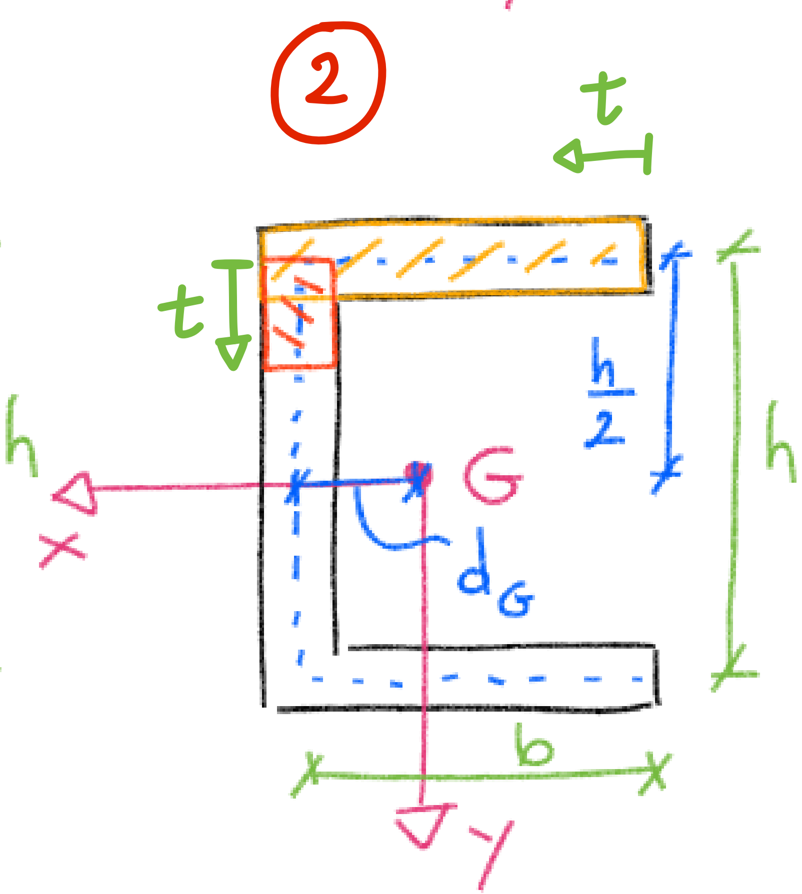
$$\tau_1 = \tau_m^{(1)}(b) = \frac{1}{2}\frac{T_y}{I_x} h b$$

2) $0 \le t \le h$: $\tau_2 = \tau_m^{(2)}\left(\frac{h}{2}\right) = \tau_1 + \frac{T_y h^2}{8 I_x}$

$$S_x^* = -\frac{h}{2} s b - \left(\frac{h}{2} - \frac{t}{2}\right) s t$$

↳ $\tau_m^{(2)}(t) = \underbrace{\frac{T_y}{2 I_x s}\left[h s b + (h-t) s t\right]}_{\text{QUADRATICO}}$





SOLLECITAZIONE COMPOSTA

STUDIO DISTR. SFORZI (Hp d_G NOTO)

STUDIO TAGLIO:

1) $0 \le t \le b$: $S_x^* = -\frac{h}{2} s t$

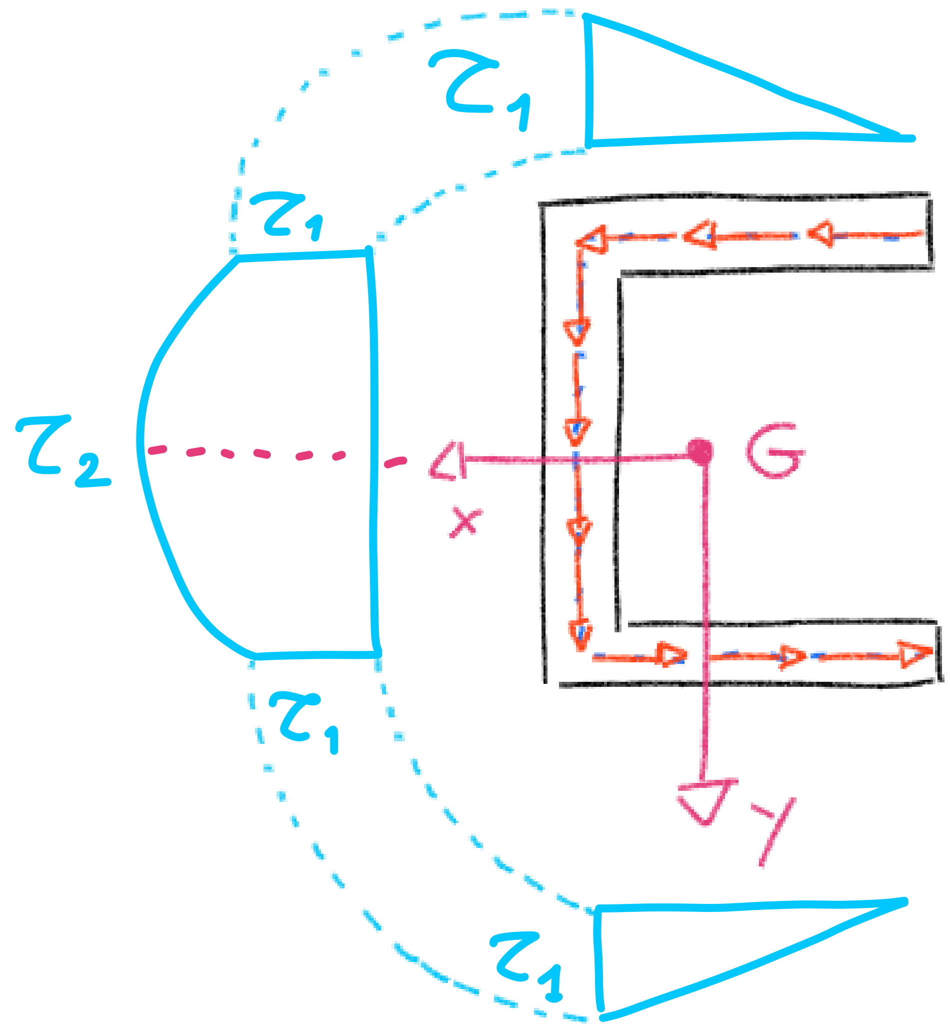
$\tau_m^{(1)}(t) = -\frac{T_y S_x^*}{I_x s} = \frac{1}{2}\frac{T_y}{I_x} h t$ } LINEARE

$\tau_1 = \tau_m^{(1)}(b) = \frac{1}{2}\frac{T_y}{I_x} h b$

2) $0 \le t \le h$: $\tau_2 = \tau_m^{(2)}\left(\frac{h}{2}\right) = \tau_1 + \frac{T_y h^2}{8 I_x}$

$S_x^* = -\frac{h}{2} s b - \left(\frac{h}{2} - \frac{t}{2}\right) s t$

$$\tau_m^{(2)}(t) = \underbrace{\frac{T_y}{2 I_x s}\left[h s b + (h-t) s t\right]}_{\text{QUADRATICO}}$$



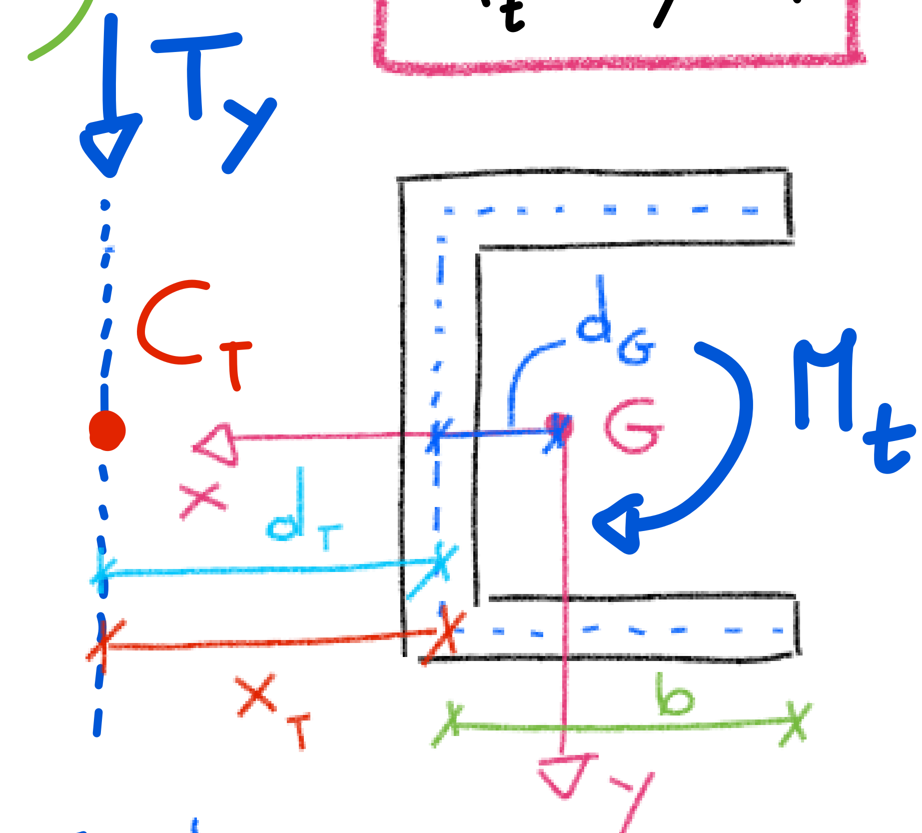
SOLLECITAZIONE COMPOSTA

STUDIO DISTR. SFORZI (Hp d_G NOTO)

$$d_G = \frac{b^2}{2b+h}; \quad I_x = \frac{sh^3}{12} + \frac{sbh^2}{2};$$

$$\boxed{x_T = d_G + d_T}$$

$$\boxed{M_t = T_y \, x_T}$$



STUDIO MOMENTO TORCENTE

- SEZIONE SOTTILE APERTA
- ANALOGIA IDRODINAMICA
- $a = 2b + h$ (linea media)

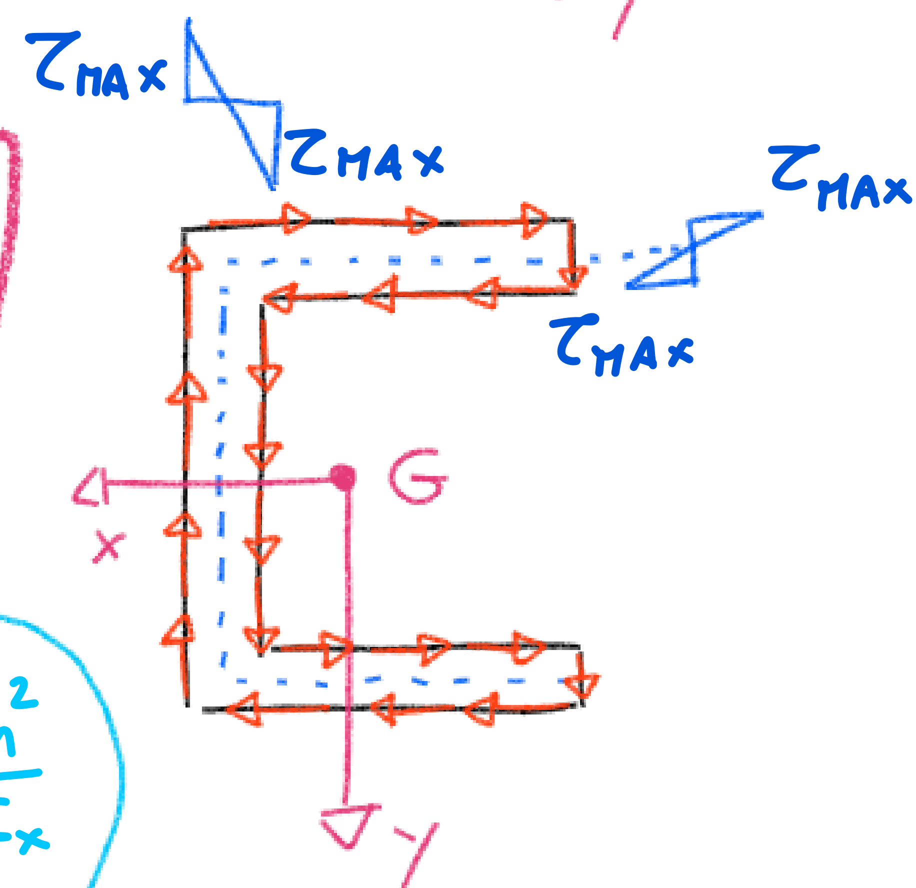
↳ $I_t = \frac{1}{3} a s^3 = \frac{1}{3}(2b+h)s^3$

- $M_t = T_y \, x_T$ (ORARIO)

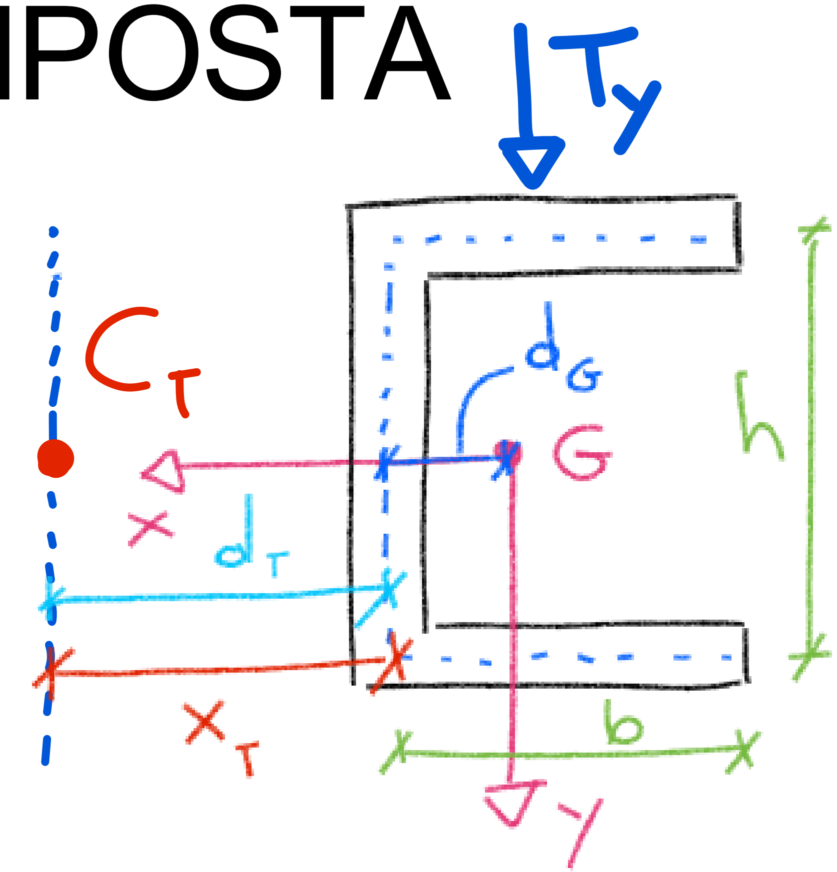
↳ $\tau_{MAX}^{(BORDI)} = \frac{M_t}{I_t} s = \frac{T_y x_T}{I_t} s$

$$\boxed{\tau^{(M_t)} \gg \tau^{(T_y)}}$$

$$\tau_2 = \tau_1 + \frac{T_y h^2}{8 I_x}$$



SOLLECITAZIONE COMPOSTA



CENTRO DI TAGLIO

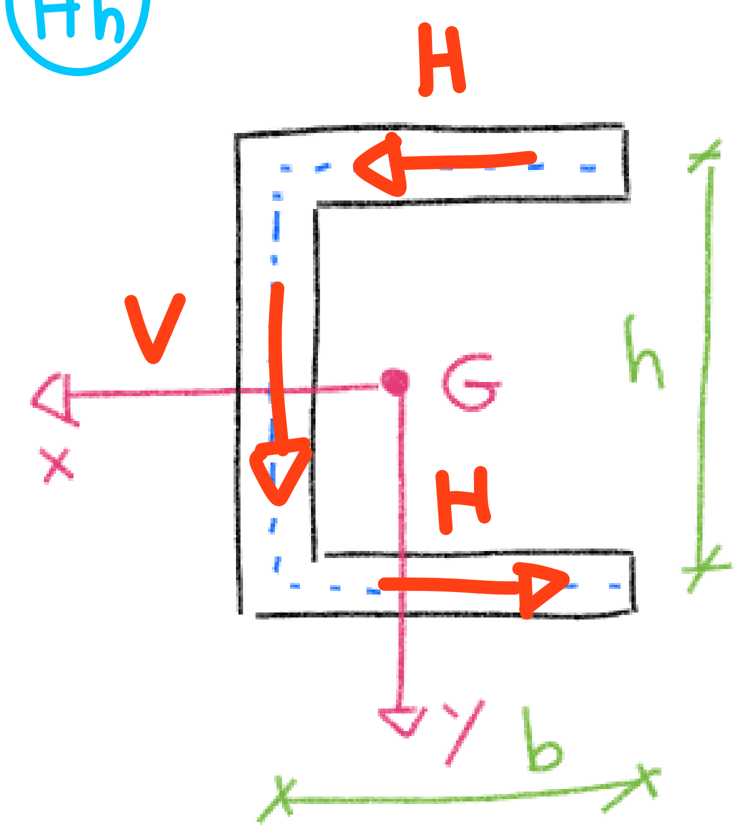
$C_T(x_T, y_T)$

PUNTO DEL PIANO, C_T, TALE CHE SE T_y PASSA PER C_T ALLORA LA SEZIONE NON RUOTA

$$d_G = \frac{b^2}{2b+h} \; ; \quad I_x = \frac{s h^3}{12} + \frac{s b h^2}{2} \; ; \quad \boxed{x_T = d_G + d_T}$$

$V = T_y$ EQUIVALENZA STATICA

$$H = \int_0^b \tau^{(T_y)} s \, dt = s \int_0^b \frac{T_y}{2 I_x} h t \, dt = \frac{T_y}{4 I_x} b^2 h s \Rightarrow Hh$$



EQUILIBRIO:

$$Hh = T_y d_T \Rightarrow \underbrace{d_T = \frac{Hh}{T_y} = \frac{b^2 h^2 s}{4 I_x} = \frac{3b^2}{6b+h}}$$

$$\underbrace{x_T = d_G + d_T = \frac{b^2}{2b+h} + \frac{3b^2}{6b+h}}$$